

Cranial Nerve Nuclei: Cranial Nerves I, II, III, IV and VI

Isaac Kurtzer, Ph.D.
Department of Biomedical Science
ikurtzer@nyit.edu

**NEW YORK INSTITUTE
OF TECHNOLOGY**

College of Osteopathic
Medicine



Do.
Make.
Heal.
Innovate.
Reinvent the Future.

Session Objectives

**NEW YORK INSTITUTE
OF TECHNOLOGY**

College of Osteopathic
Medicine

1. Identify the cranial nerves for vision, and olfaction, and eye behavior.
2. Identify the cranial nerve nuclei for eye behavior and its coordination.
3. Locate injuries to these pathways given particular deficits and residual capabilities.

Nolte's The Human Brain: An introduction to its functional anatomy

Chapter 12 Cranial Nerves and their nuclei

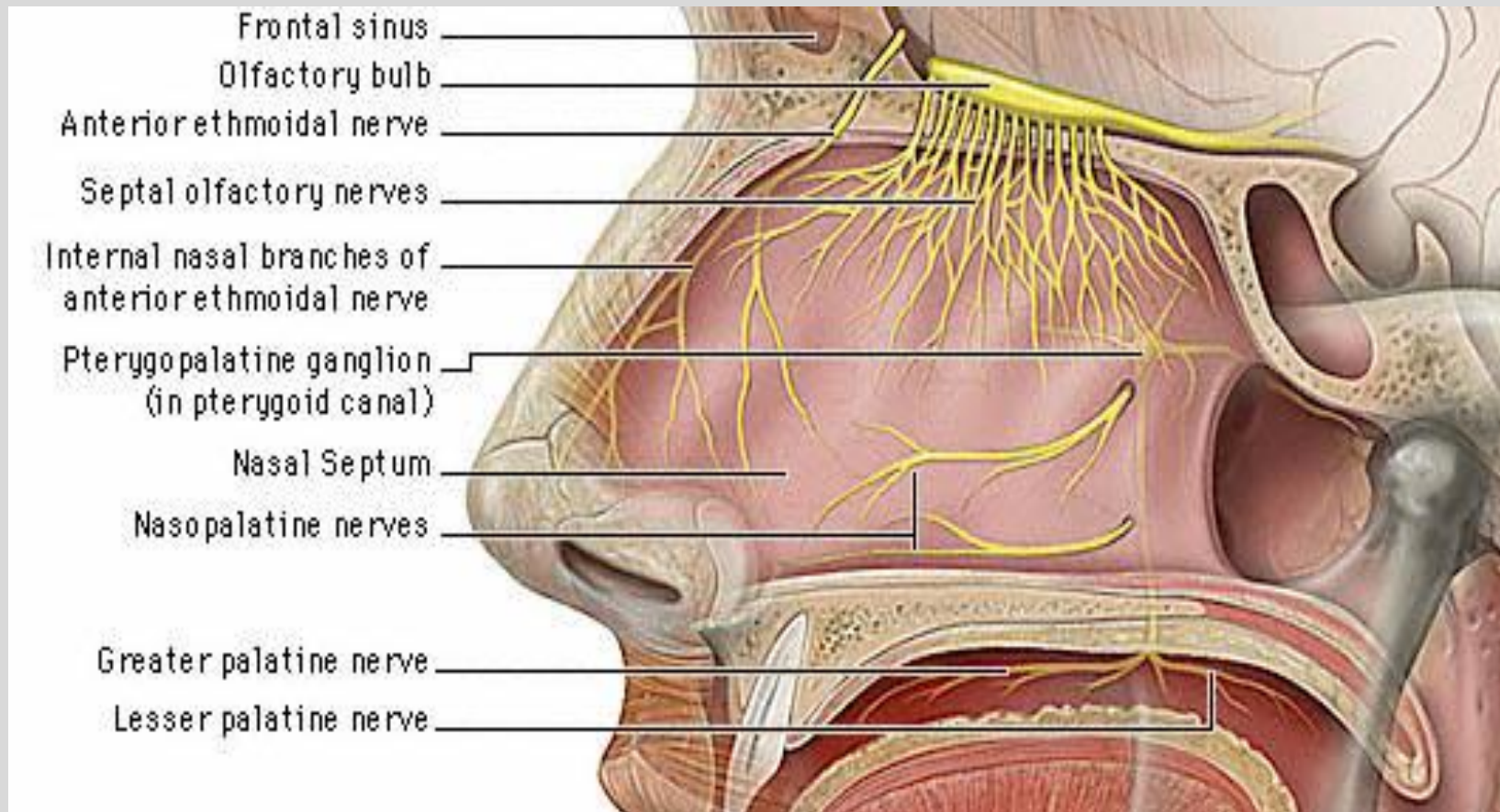
<http://arktos.nyit.edu/login?url=https://www-clinicalkey-com.arktos.nyit.edu/#!/content/book/3-s2.0-B9780323653985000126>

CN I (olfactory nerve)

Smell is VISCERAL SENSORY

olfactory epithelium cells → *olfactory bulb* → olfactory tract

the only sensory fibers to bypass thalamus to cerebral cortex



CN I (olfaction) dysfunction

Anosmia – loss of olfactory sensation (can be ipsilateral, therefore, subtle)

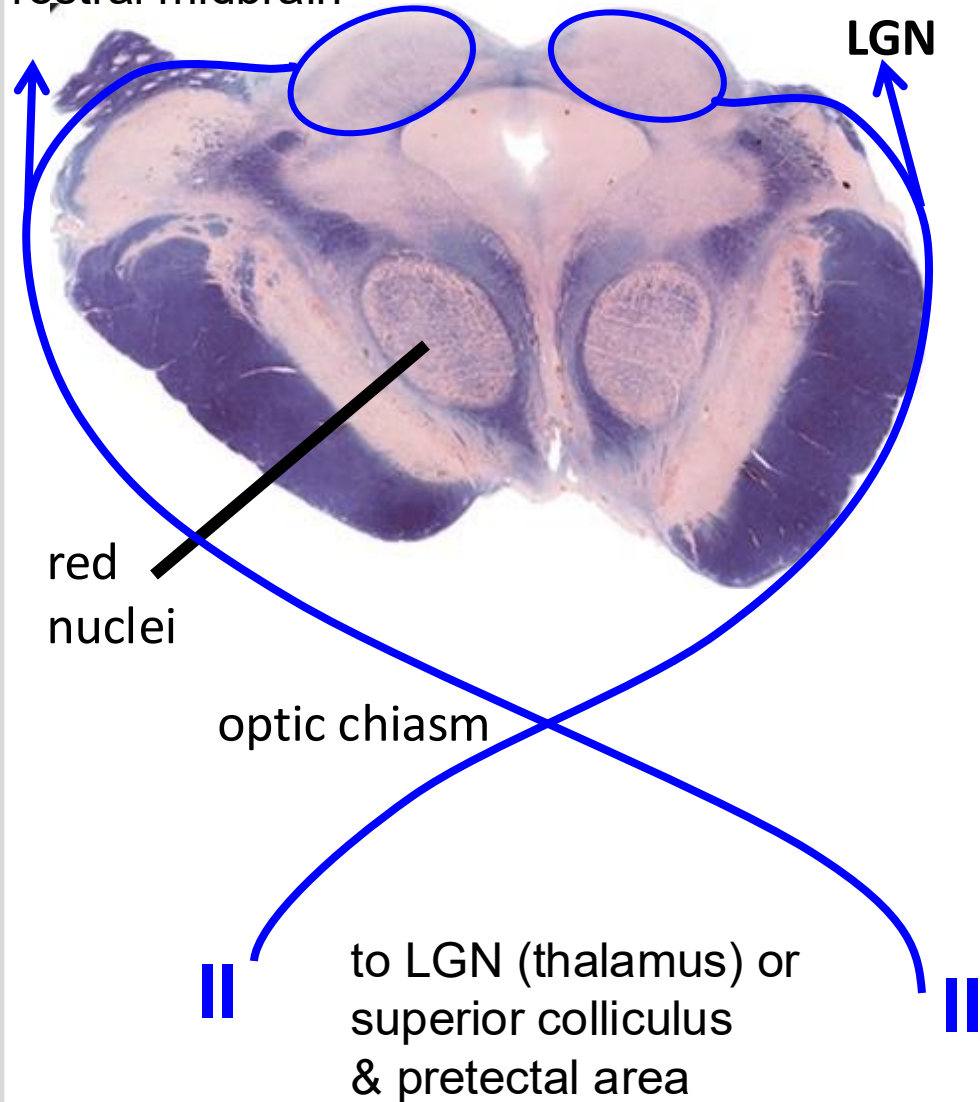
Excessive head movement (impact)
Pulling or shearing of olfactory fibers

Frontal lobe tumors or abscesses
Olfactory tract compression

Early sign of Parkinson's disease
Lewy bodies present in olfactory bulbs

CN II (optic)

rostral midbrain



NEW YORK INSTITUTE OF TECHNOLOGY

College of Osteopathic
Medicine

Vision is SPECIAL SENSORY

retina → optic nerves - optic chiasm (some decussate) –
optic tract
→ lateral geniculate nucleus of thalamus → occipital lobe

Or

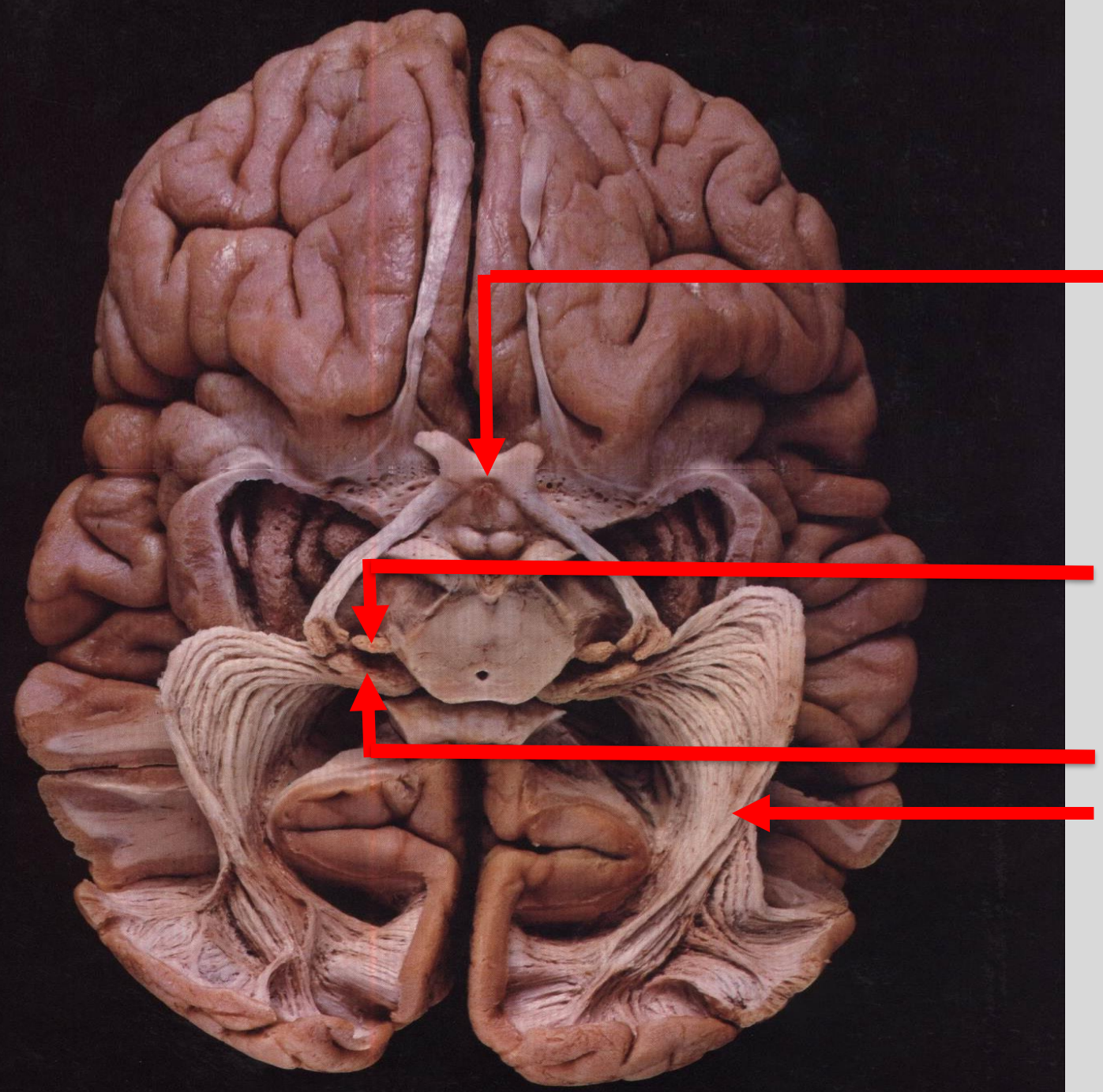
→ superior colliculus

Pupillary light reflex is VISCERAL FEEDBACK

retina → optic nerves - optic chiasm - optic tract →
pretectal area → (bilaterally to) Edinger Westphal nuclei

NEW YORK INSTITUTE OF TECHNOLOGY

College of Osteopathic
Medicine



optic
chiasm

to
superior colliculus and
pretectal area

to
thalamus

to
visual
cortex

CN II (vision) dysfunction

Trauma

poorly protected by eyelid

Vascular

diabetes

glaucoma

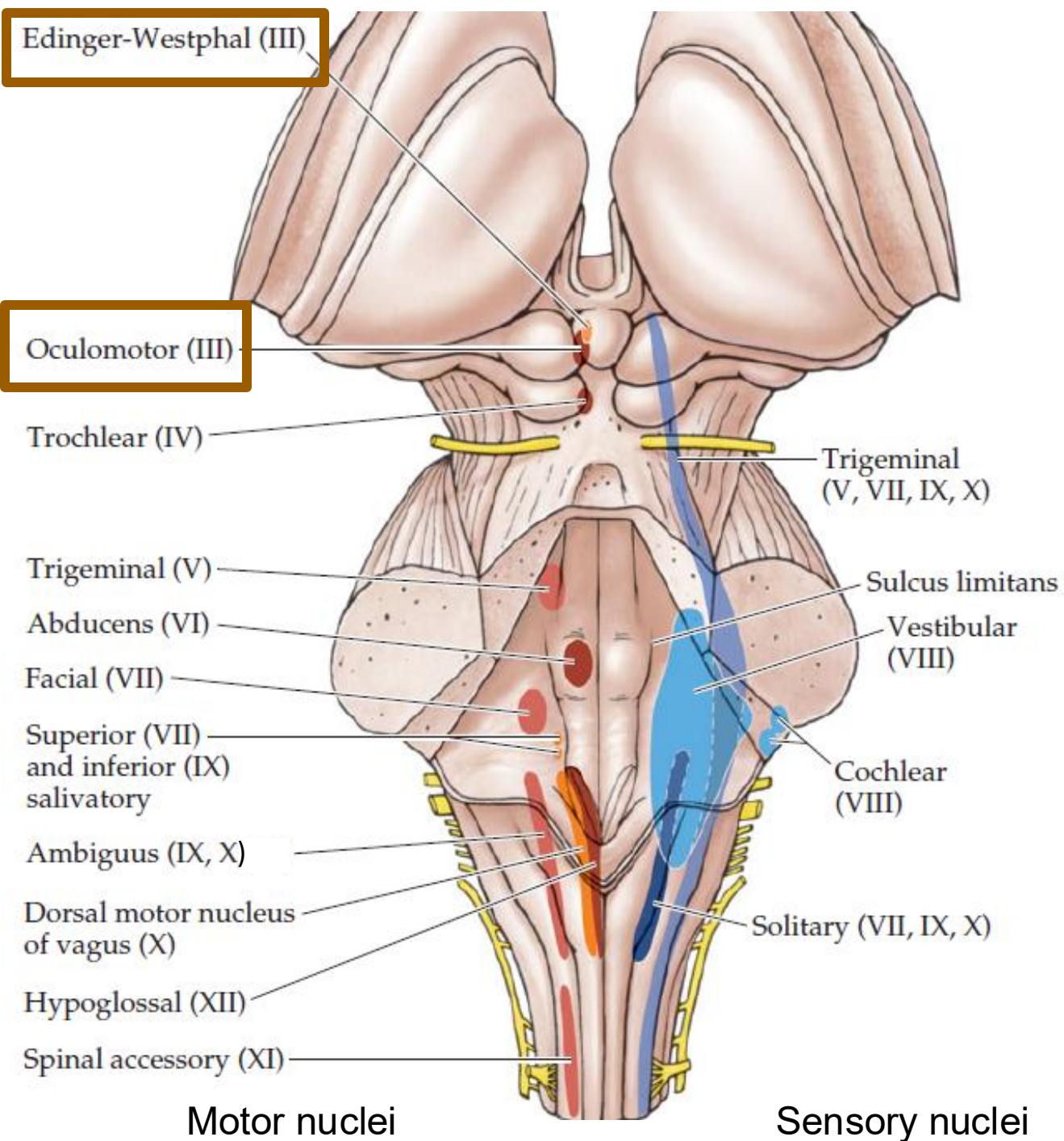
**Impacted by many conditions and
reflective of many conditions**

CNS Disease

multiple sclerosis

CNS tumors

	<u>SOMATIC</u> <u>MOTOR</u>	<u>BRANCHIAL</u> <u>MOTOR</u>	<u>VISCERAL</u> <u>MOTOR</u>	<u>VISCERAL</u> <u>SENSORY</u>	<u>SOMATIC</u> <u>SENSORY</u>	<u>SPECIAL</u> <u>SENSORY</u>
<i>III</i>	<i>oculomotor</i> → eye muscles		<i>Edinger</i> <i>Westphal</i> → iris			
<i>IV</i>	<i>trochlear</i> → eye muscles					
<i>VI</i>	<i>abducens</i> → eye muscles					

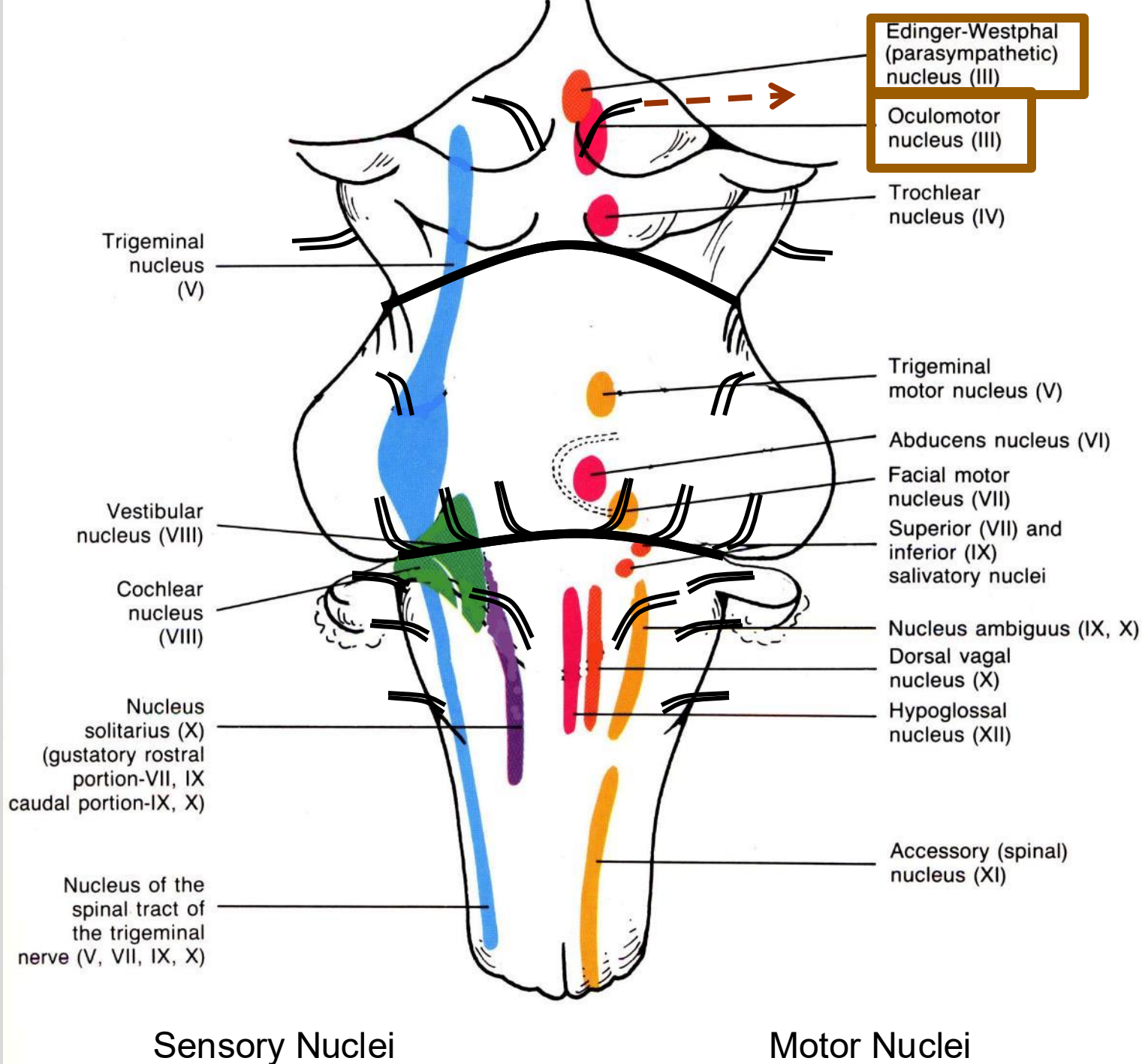


NEW YORK INSTITUTE OF TECHNOLOGY

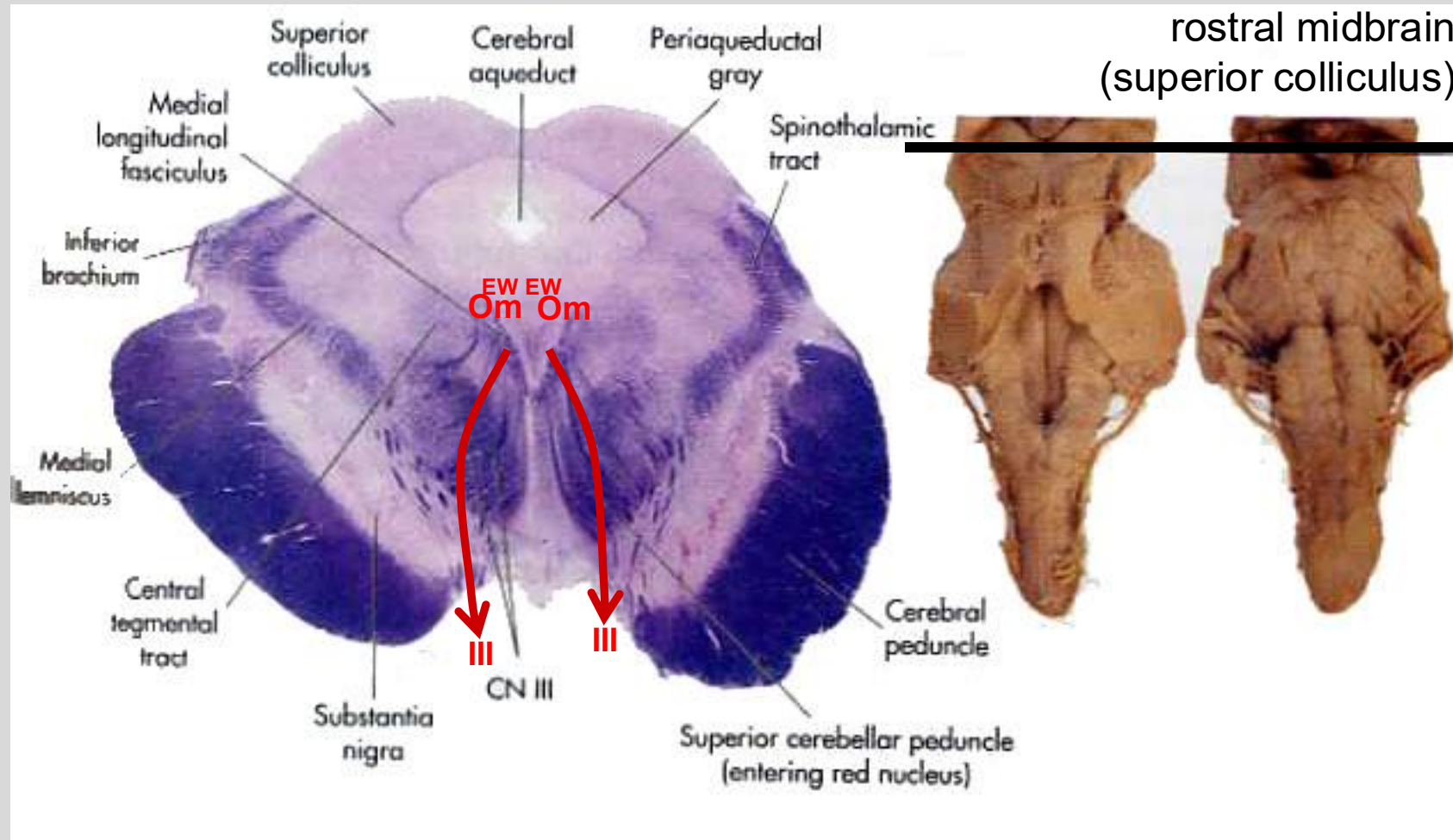
College of Osteopathic
Medicine

CN III (oculomotor)

CN III (oculomotor)



CN III (oculomotor)



III	Edinger Westphal	lens & pupil (smooth muscle)
III	oculomotor	eye muscles (striated)

SOMATIC MOTOR : ocular positioning and eyelid *oculomotor nucleus* →

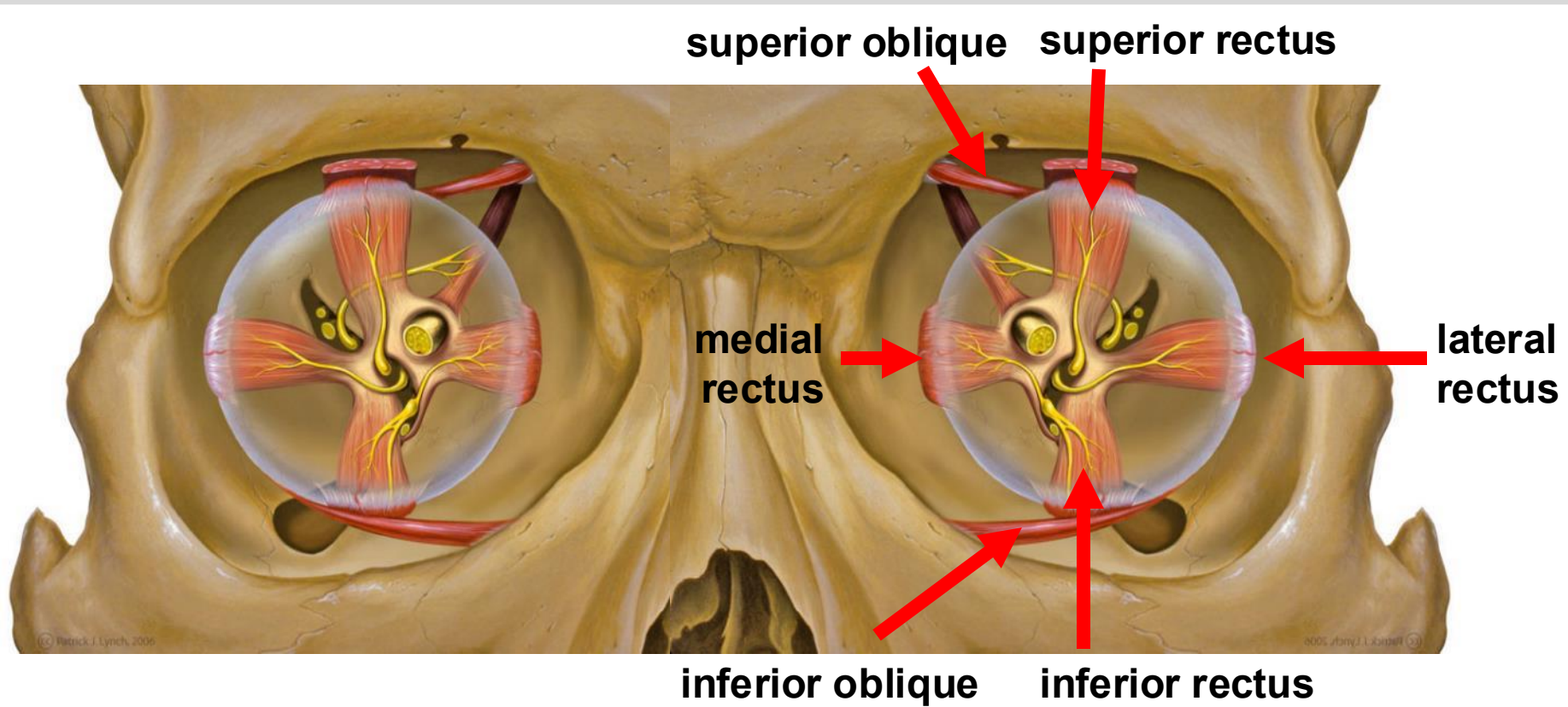
eyes: medial rectus (ipsilateral) - adduction
inferior rectus (ipsilateral) – mainly depression
inferior oblique (ipsilateral) - extorsion (outward rotation)
superior rectus (**contralateral**) – mainly elevation

eyelid: levator palpebrae superioris (**contralateral**)

NEW YORK INSTITUTE
OF TECHNOLOGY

College of Osteopathic
Medicine

CN III (oculomotor)



SOMATIC MOTOR : ocular positioning and eyelid *oculomotor nucleus* →

eyes: medial rectus (ipsilateral) - adduction

inferior rectus (ipsilateral) – mainly depression

inferior oblique (ipsilateral) - extorsion (outward rotation)

superior rectus (**contralateral**) – mainly elevation

eyelid: levator palpebrae superioris (**contralateral**)

NEW YORK INSTITUTE
OF TECHNOLOGY

College of Osteopathic
Medicine

CN III (oculomotor)

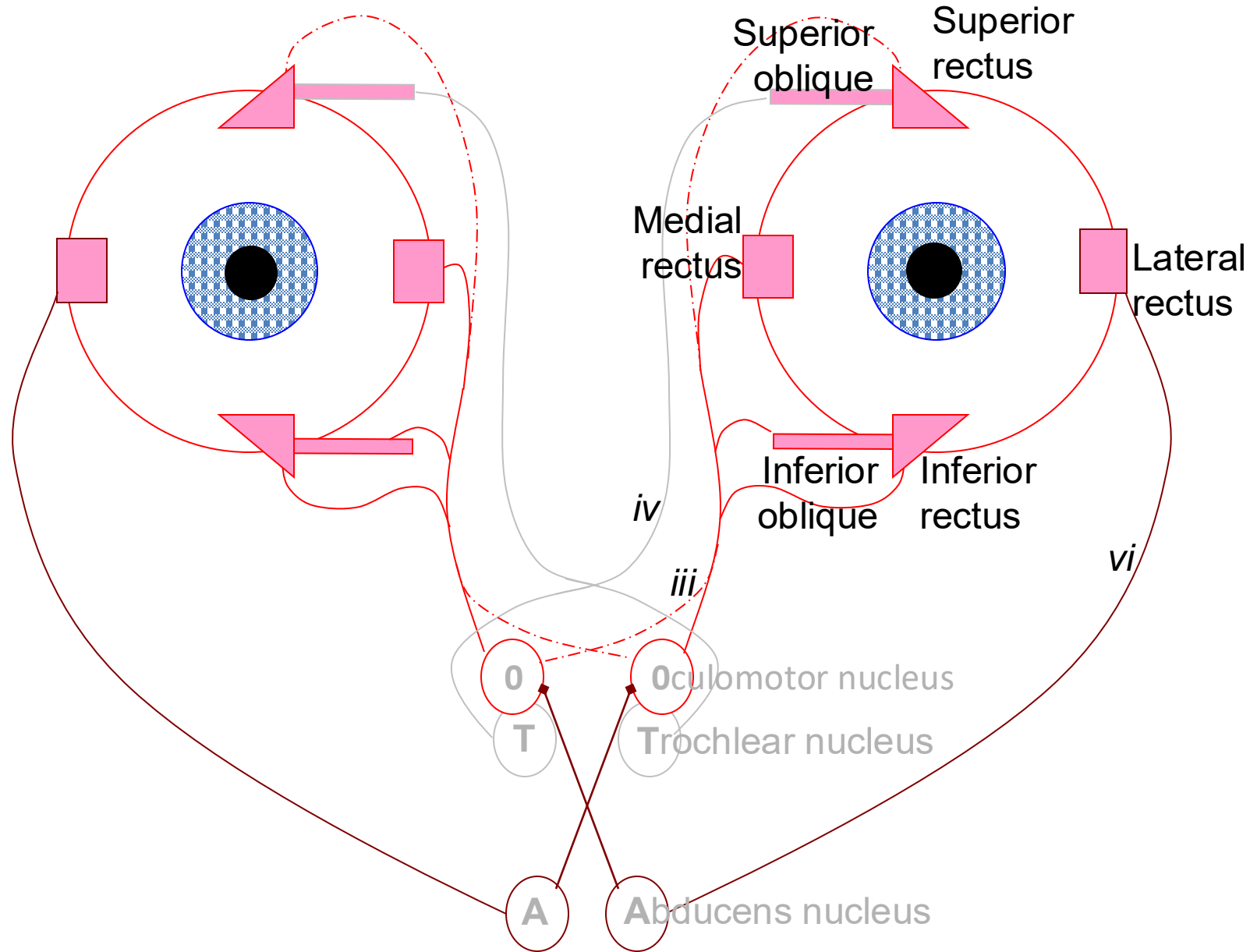
VISCERAL EFFERENT : light reflex

Edinger Westphal nuclei (part of *oculomotor complex*) →

pupil contraction → less light: sphincter pupillae (ipsilateral)

lens accommodation * → more convex / shorten focal length: ciliary muscle (ipsilateral)

CN III includes oculomotor nuclei efferents of both sides



Remember - the nerves always control the ipsilateral side

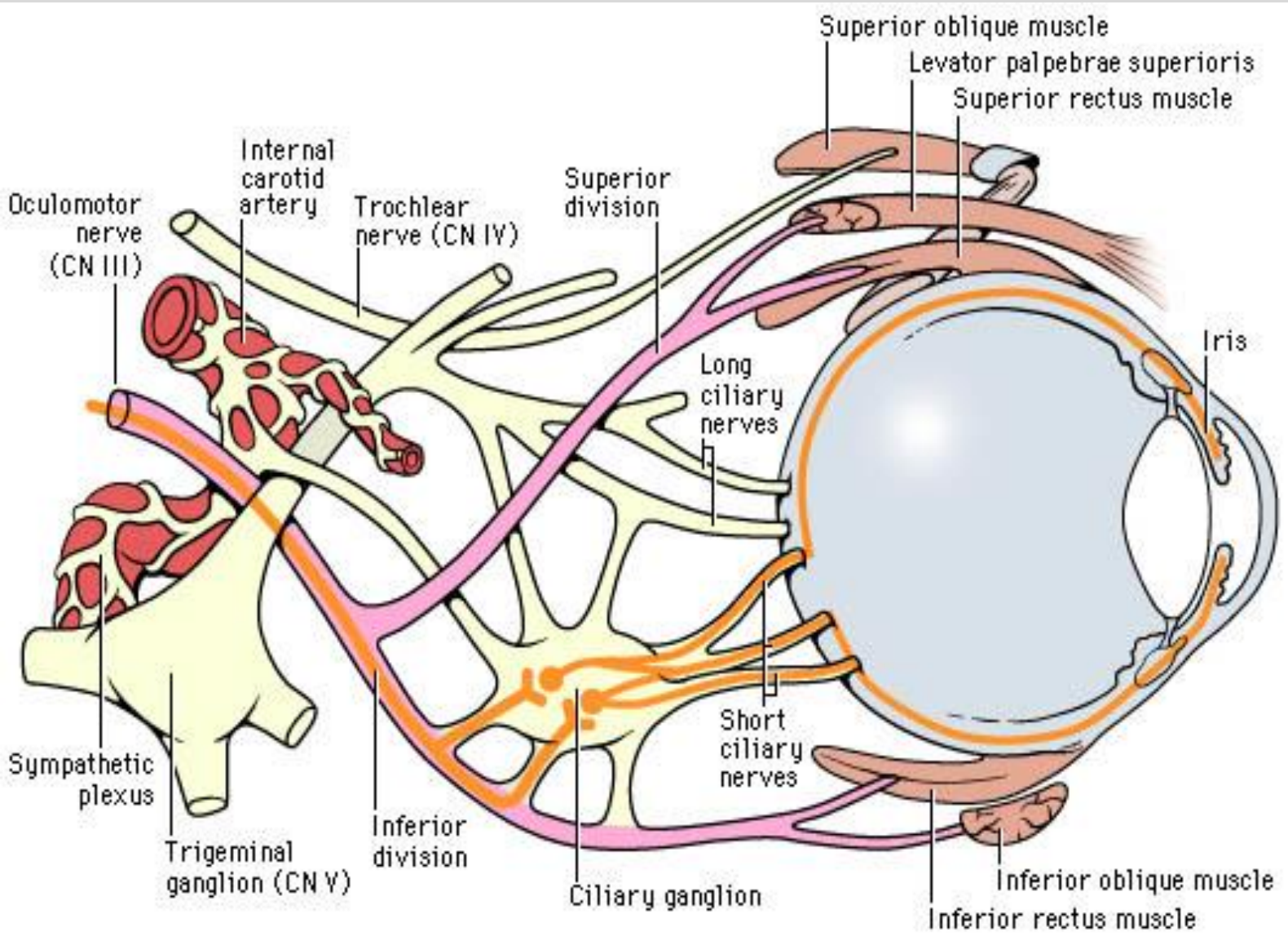
**NEW YORK INSTITUTE
OF TECHNOLOGY**

College of Osteopathic
Medicine

CN III (oculomotor): superior and inferior divisions

NEW YORK INSTITUTE
OF TECHNOLOGY

College of Osteopathic
Medicine



contralateral oculomotor nucleus

superior division

superior rectus &
levator palpebrae superioris

ipsilateral oculomotor and EW nucleus

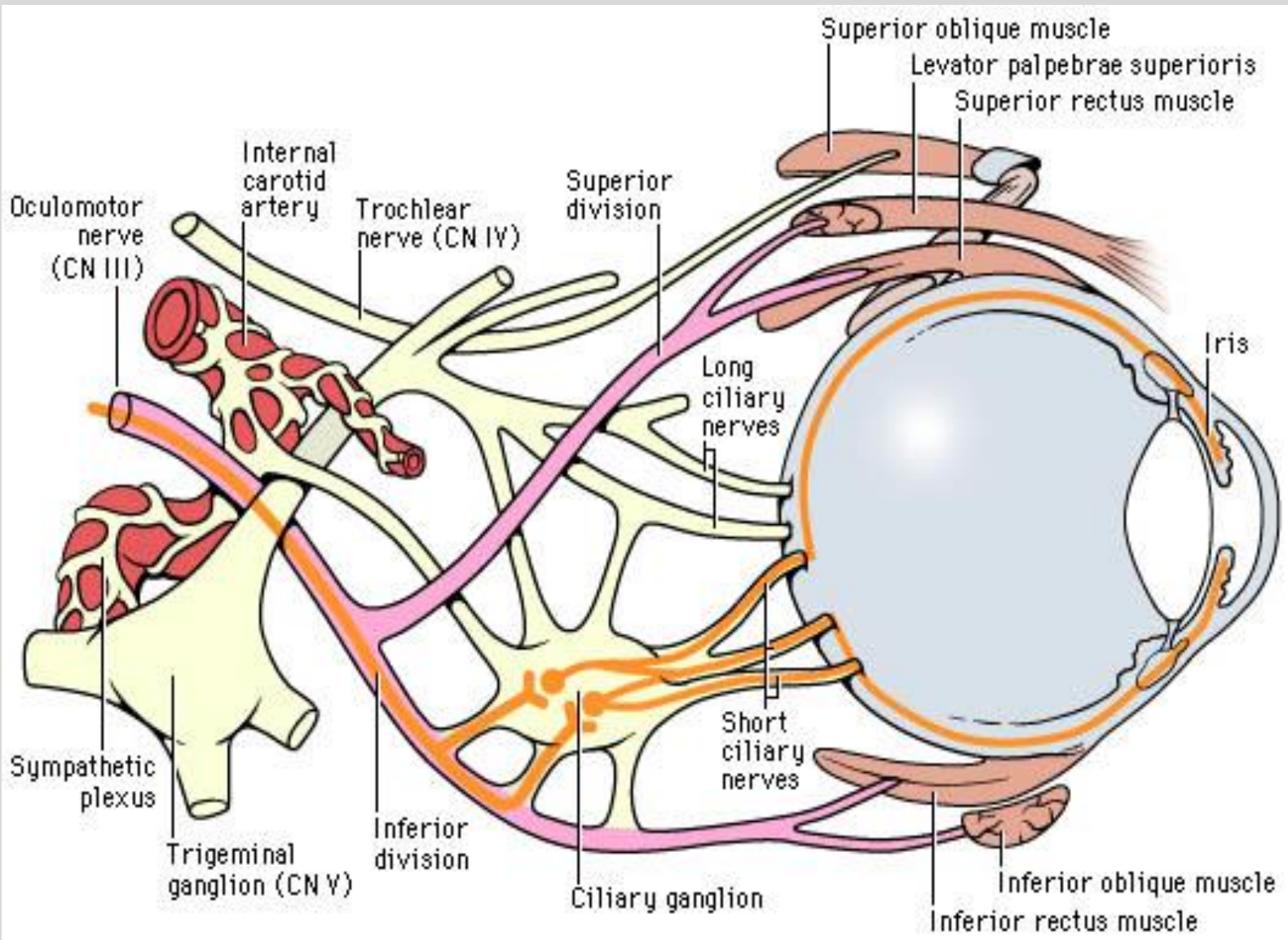
inferior division

medial rectus,
inferior rectus,
inferior oblique, and
parasympathetics

CN III (oculomotor): brainstem or CN lesion?

**NEW YORK INSTITUTE
OF TECHNOLOGY**

College of Osteopathic
Medicine



Unilateral brainstem lesions have ipsilateral lower branch symptoms and contralateral upper branch symptoms

Unilateral CN lesions only have ipsilateral symptoms

CN III palsy

Ptosis



weak or paralyzed levator
palpebrae superioris
*lesion located in the
superior division*

Medial rectus palsy



*lesion located in
inferior division*

Complete lesion



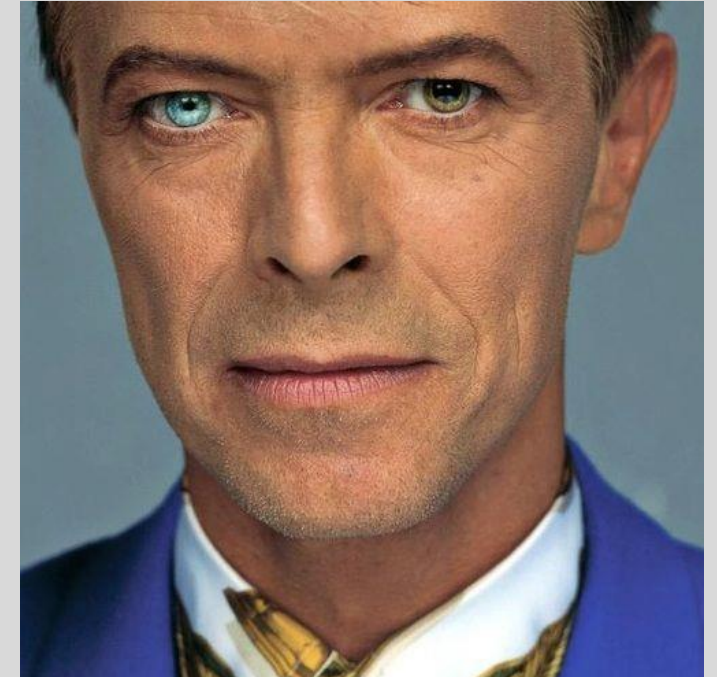
weakness or paralysis of 4 of
the 6 extraocular muscles.
Unopposed muscles pull eye
down and out

**NEW YORK INSTITUTE
OF TECHNOLOGY**

College of Osteopathic
Medicine

Anisocoria

David Bowie!

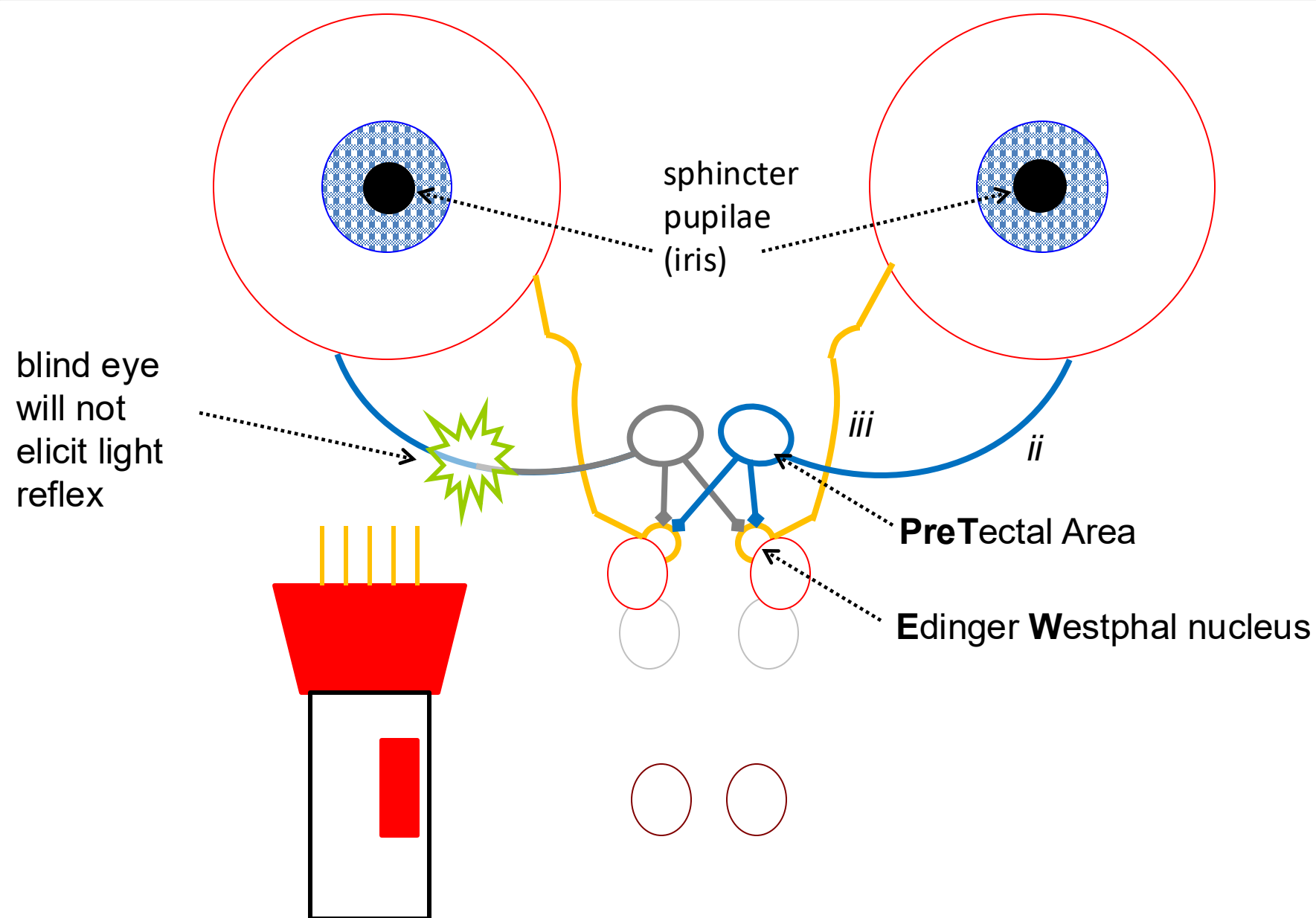


Weakness or paralysis of sphincter pupillae. Affected
eye remains more dilated than contralateral eye.

*lesion located in
inferior division*

Light reflex

retina → CN II → PreT → bilateral EW nuclei → each CN III → each iris

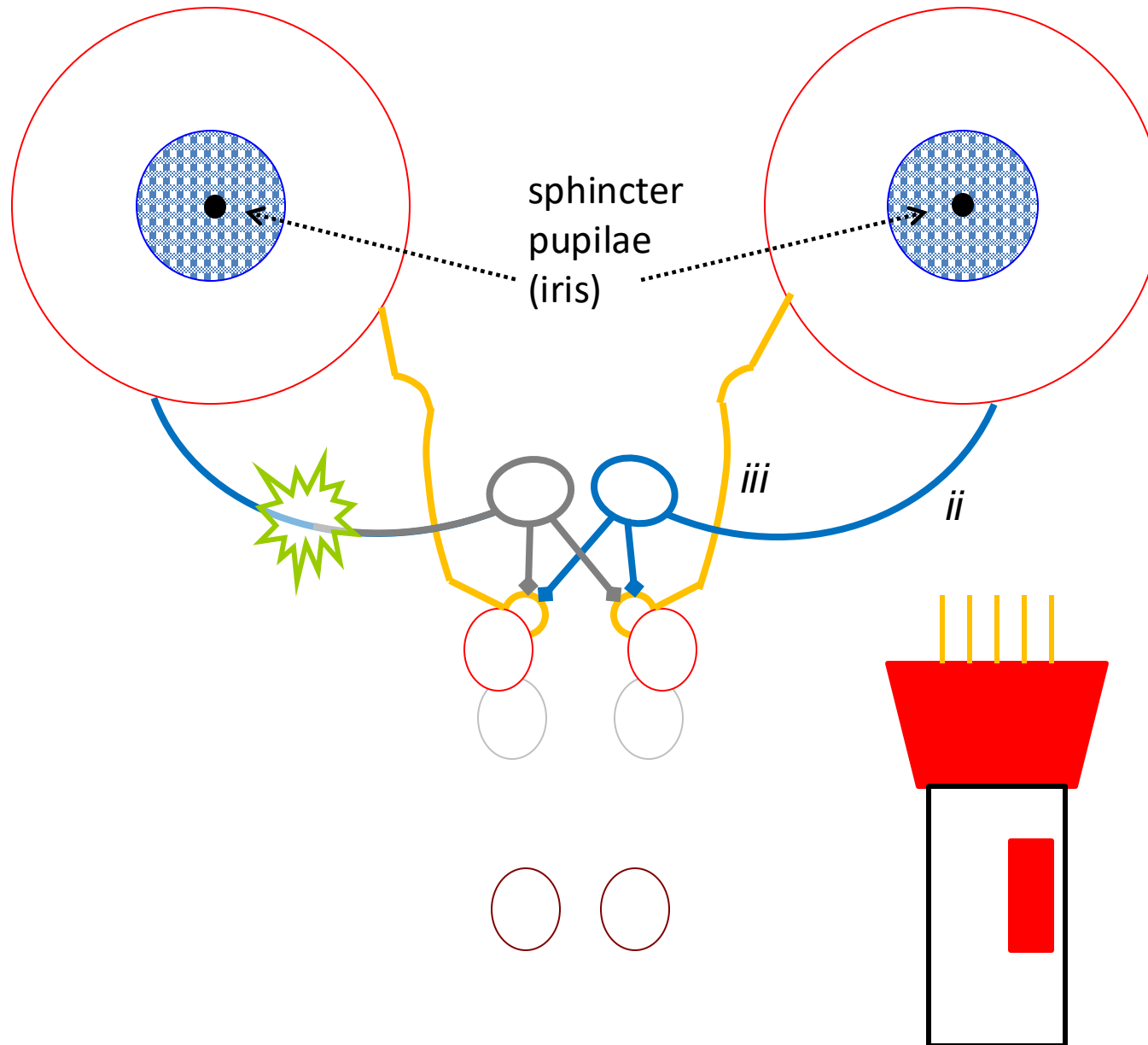


**NEW YORK INSTITUTE
OF TECHNOLOGY**

College of Osteopathic
Medicine

Light reflex

retina → CN II → PreT → **bilateral** EW nuclei → **each** CN III → **each** iris



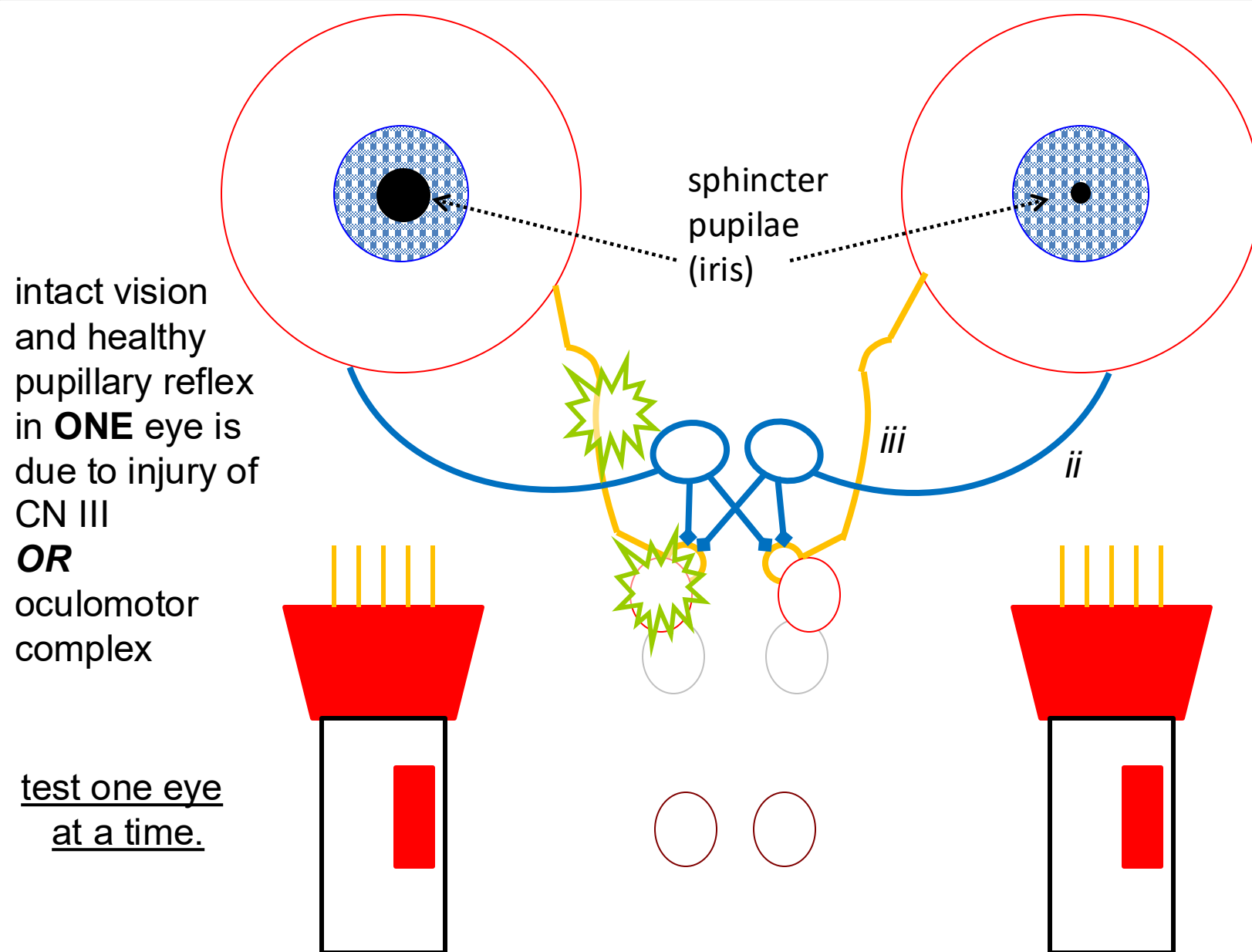
blind eye
will not
elicit light
reflex,
but can
respond to
light in the
other eye!

**NEW YORK INSTITUTE
OF TECHNOLOGY**

College of Osteopathic
Medicine

Light reflex

retina → CN II → PreT → bilateral EW nuclei → **each** CN III → **each** iris



**NEW YORK INSTITUTE
OF TECHNOLOGY**

College of Osteopathic
Medicine

CN III dysfunction

CN III compression - from head injury

Vascular problems

Aneurisms of posterior cerebral or superior cerebellar arteries (CN III emerges in between)

Basal midbrain infarction

Inflammation

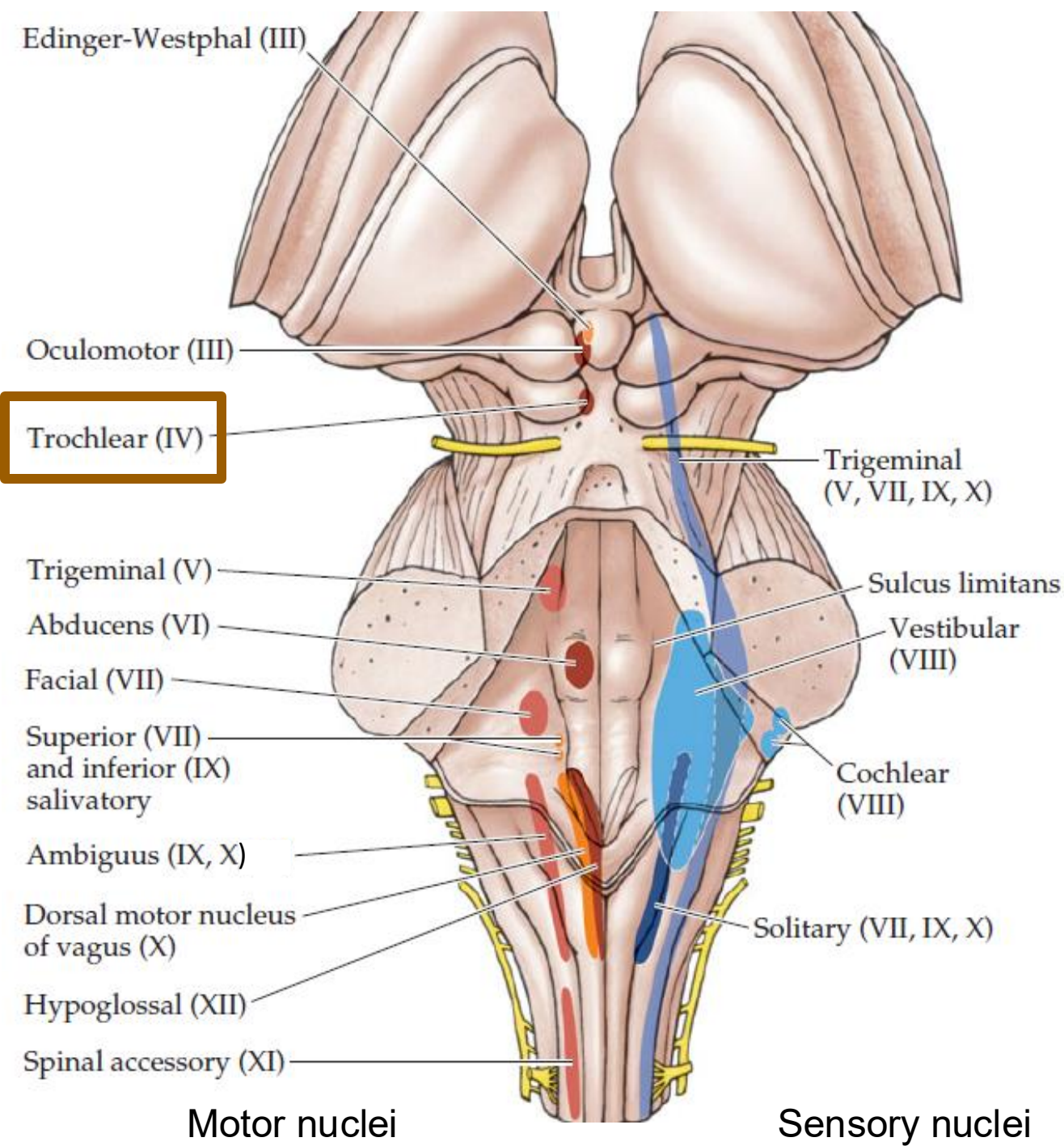
Syphilitic & tuberculous meningitis

Often Interpeduncular fossa – optic chiasm

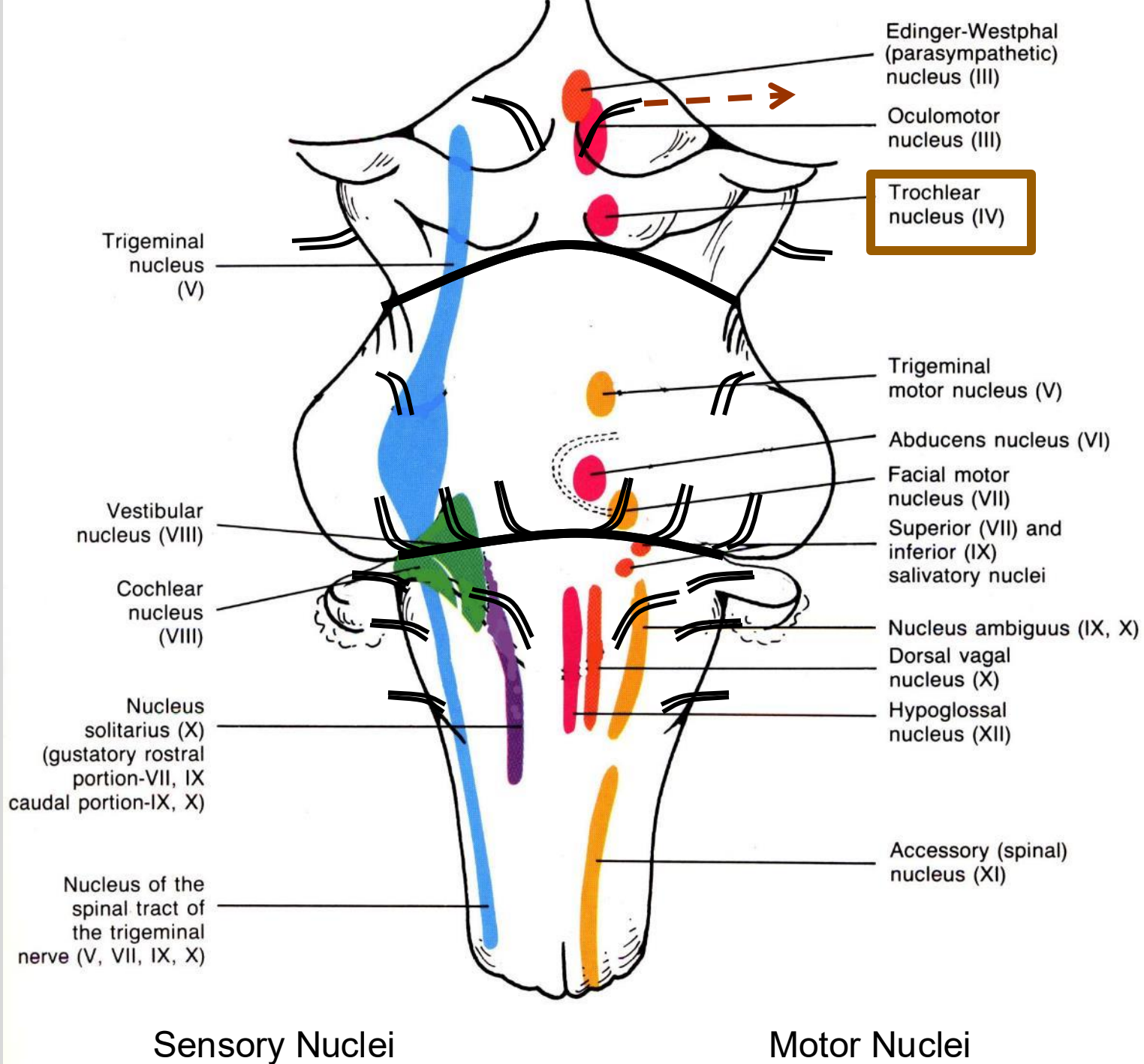
Tumor, abscess, trauma

Temporal lobe displaces CN III

CN IV (trochlear)



CN IV (trochlear)



CN IV (trochlear)

SOMATIC MOTOR : ocular positioning

***trochlear nuclei* →**

eyes:

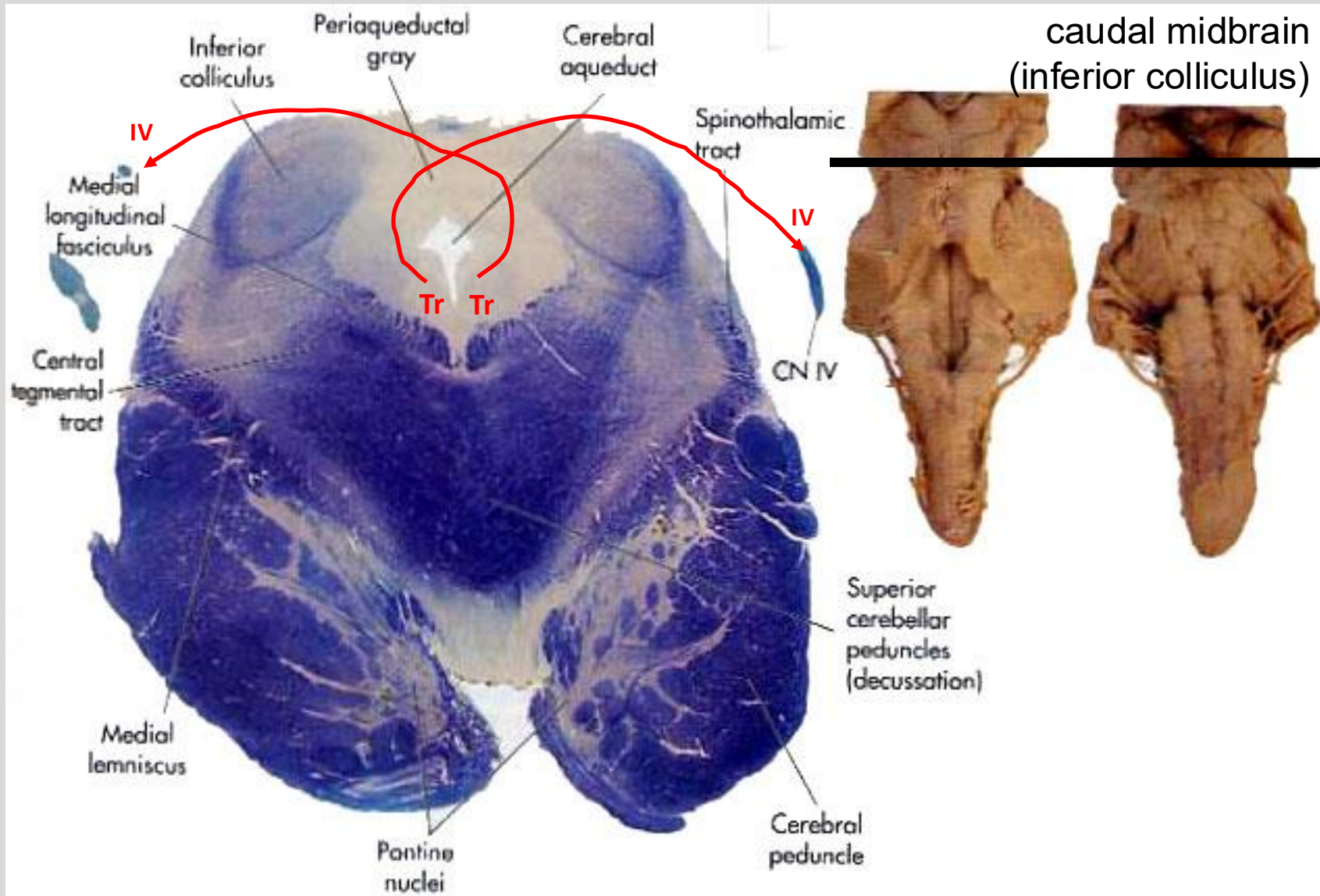
superior obliques - intortion (inward rotation)

The only nerve leaving the brainstem dorsally and has a tract that decussates

- all of the lower motor axons decussate within the brain

The smallest CN ~2400 axons (CNII ~1 Mio)

Longest intracranial course (7.5cm)

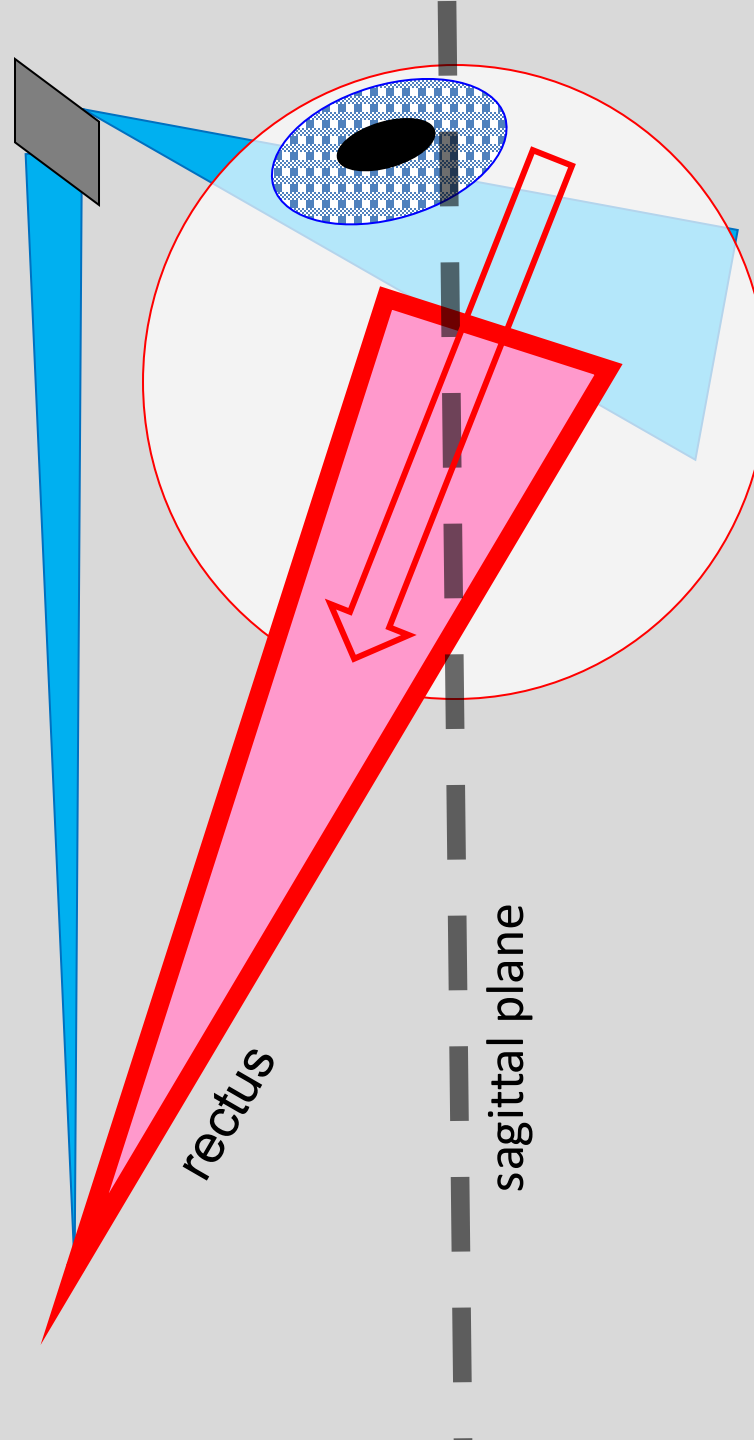
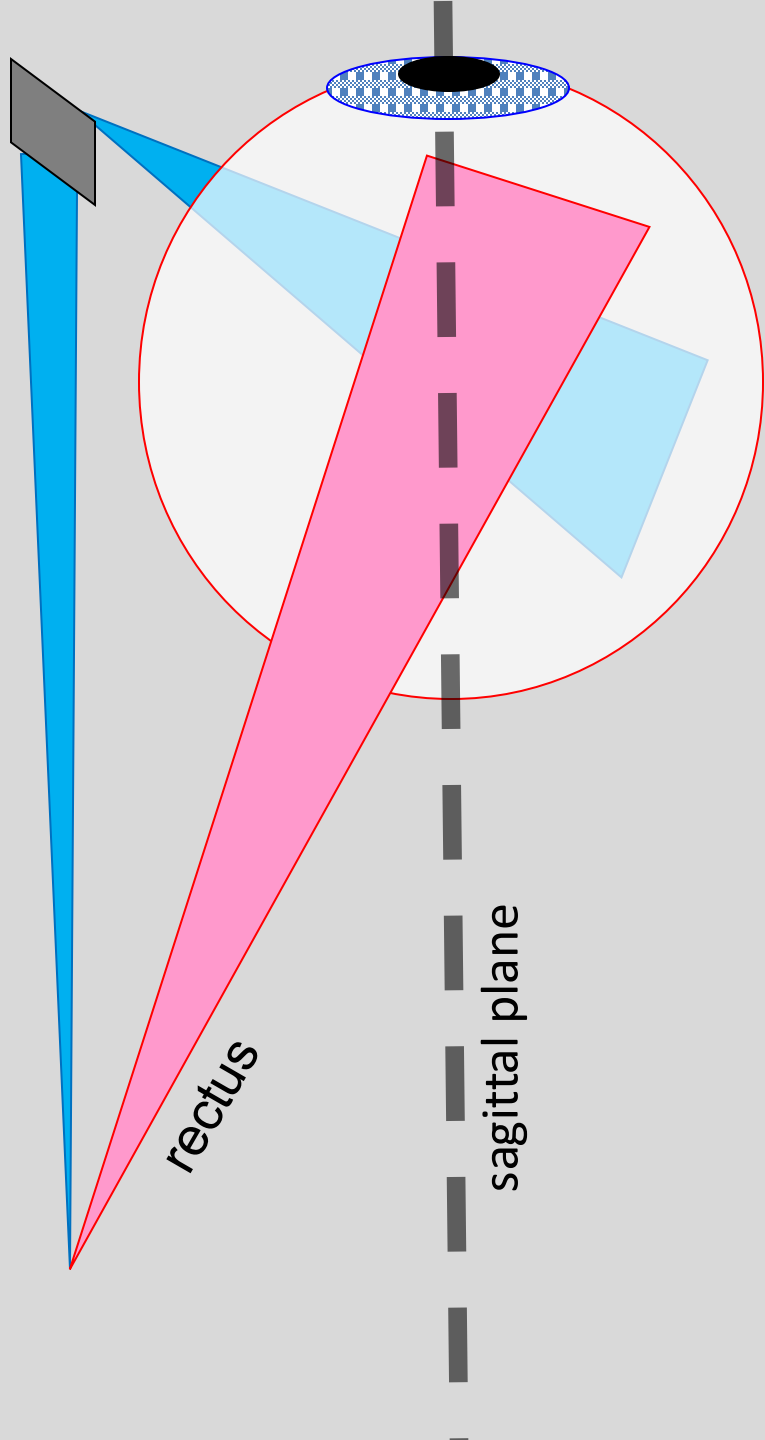


NEW YORK INSTITUTE OF TECHNOLOGY

College of Osteopathic
Medicine

CN IV (trochlear)

IV trochlear eyes muscles



NEW YORK INSTITUTE OF TECHNOLOGY

College of Osteopathic
Medicine

superior and inferior
rectus muscles attach
to orbit **out** of eye's
midsagittal plane

rectus muscles rotate
pupil **out** of eye's
midsagittal plane

CN IV (trochlear) dysfunction

Upward and inward deviation of the eye

often unconsciously compensate with head tilt to opposite shoulder, looks like torticollis



Vertical Diplopia (double vision)

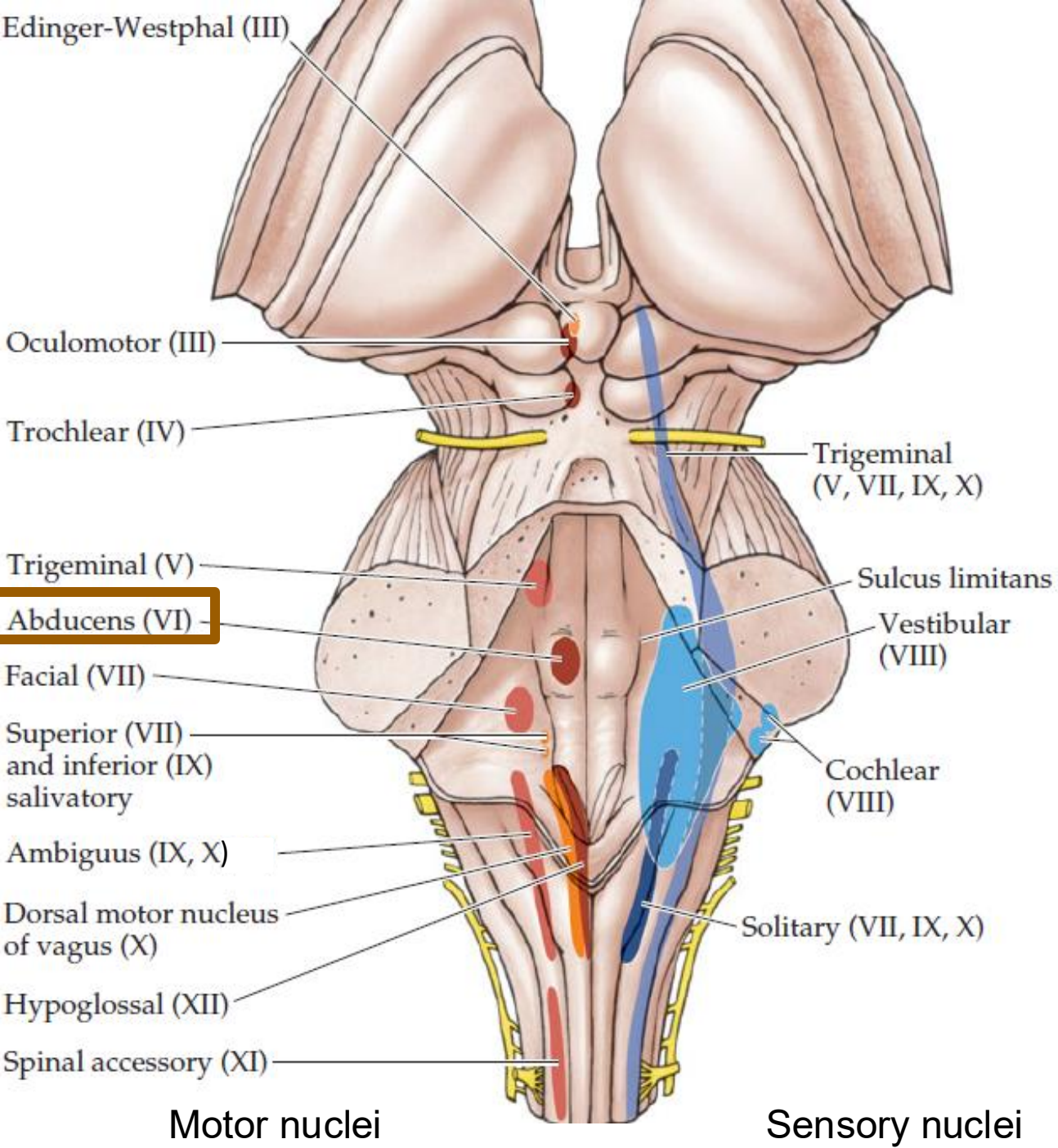
worse when looking down

CN IV is susceptible to swelling and injury along its long course
Many/most cases are congenital.

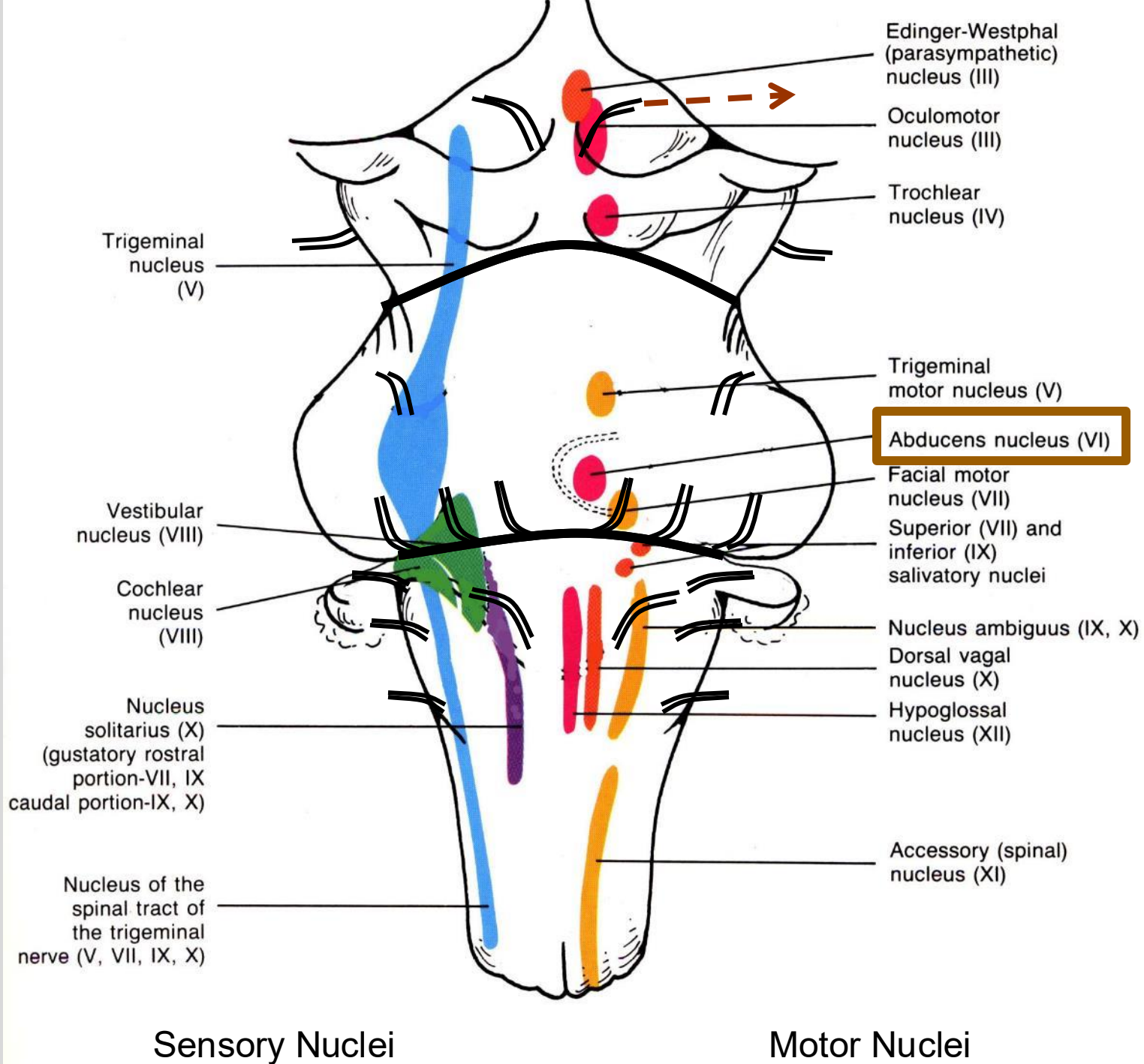
**NEW YORK INSTITUTE
OF TECHNOLOGY**

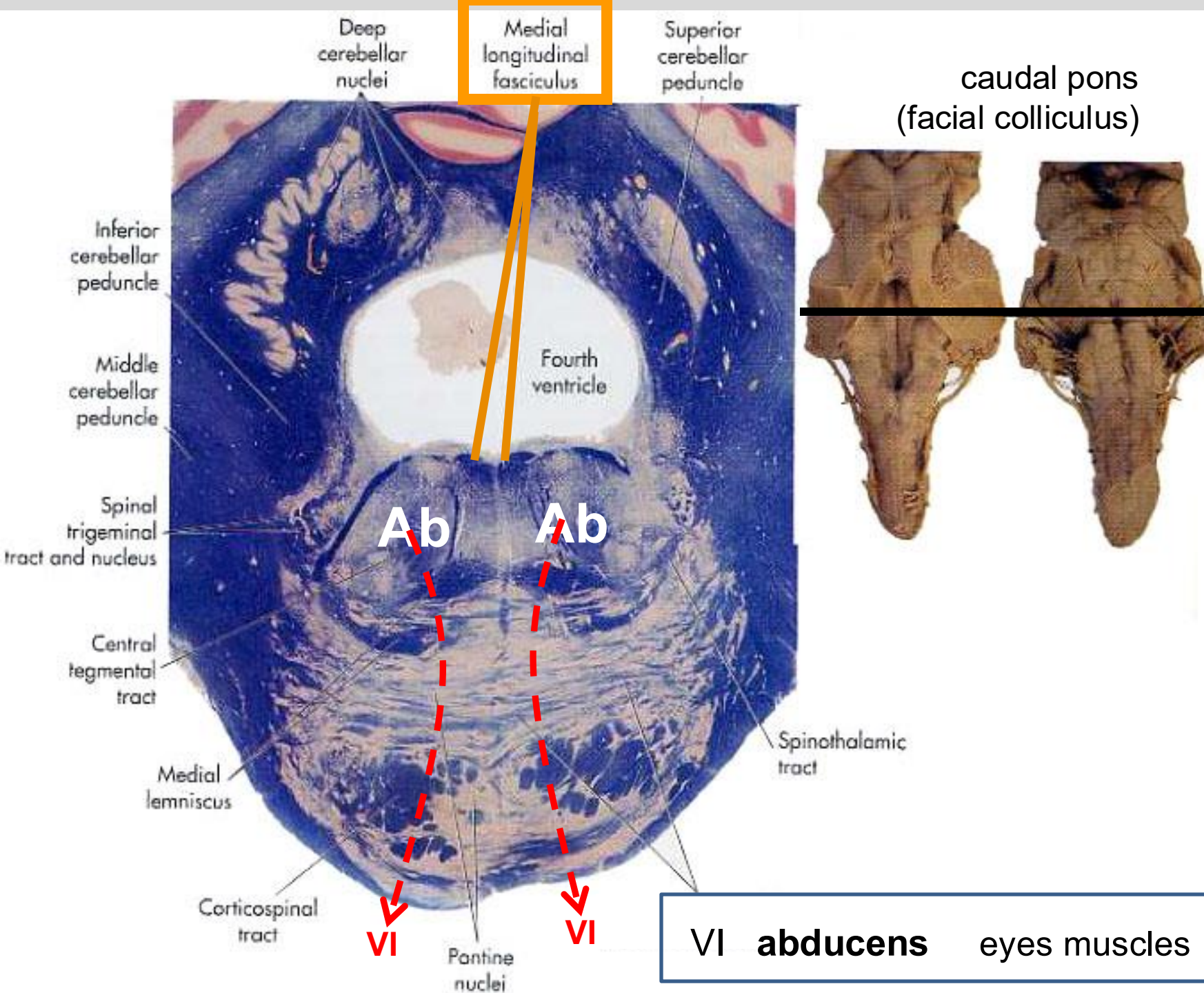
College of Osteopathic
Medicine

CN VI (abducens)



CN VI (abducens)





NEW YORK INSTITUTE OF TECHNOLOGY

College of Osteopathic
Medicine

CN VI (abducens)

CN VI (abducens)

SOMATIC MOTOR : ocular positioning

abducens nuclei →

eyes:

lateral rectus: abduction (away from midline)

CN VI paralysis is the most common isolated muscle palsy

Wide range of causes including trauma and diabetic neuropathy

Affected eye pulls medially “esotropia”, eso → inner

Horizontal Diplopia (side-by-side double vision)

Treatment

Glasses correcting refractive error (light path)

Eye muscle surgery



CN VI (abducens)

SOMATIC MOTOR : ocular positioning

abducens nuclei →

eyes:

lateral rectus: abduction (away from midline)

ALSO → contralateral oculomotor nucleus via
Medial Longitudinal Fasciculus (not CN VI)
to coordinate conjugate lateral gaze
(both eyes in same external direction)

MLF bilateral tracts travel between upper
medulla to midbrain. Decussations occur
immediately before entering.

Impacted by multiple sclerosis, brainstem
strokes, head trauma and Arnold-Chiari
malformations.

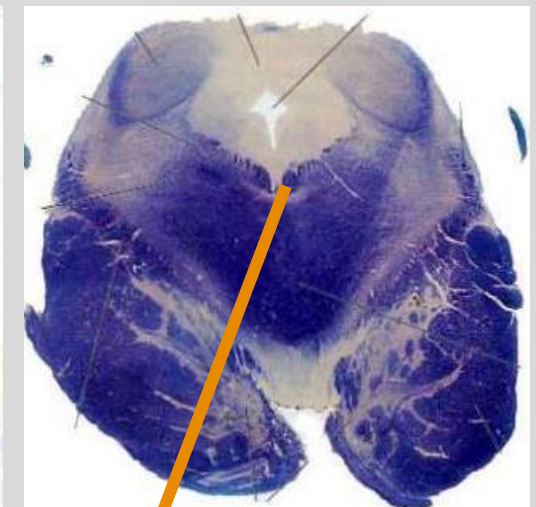
**NEW YORK INSTITUTE
OF TECHNOLOGY**

College of Osteopathic
Medicine

Pons



Midbrain



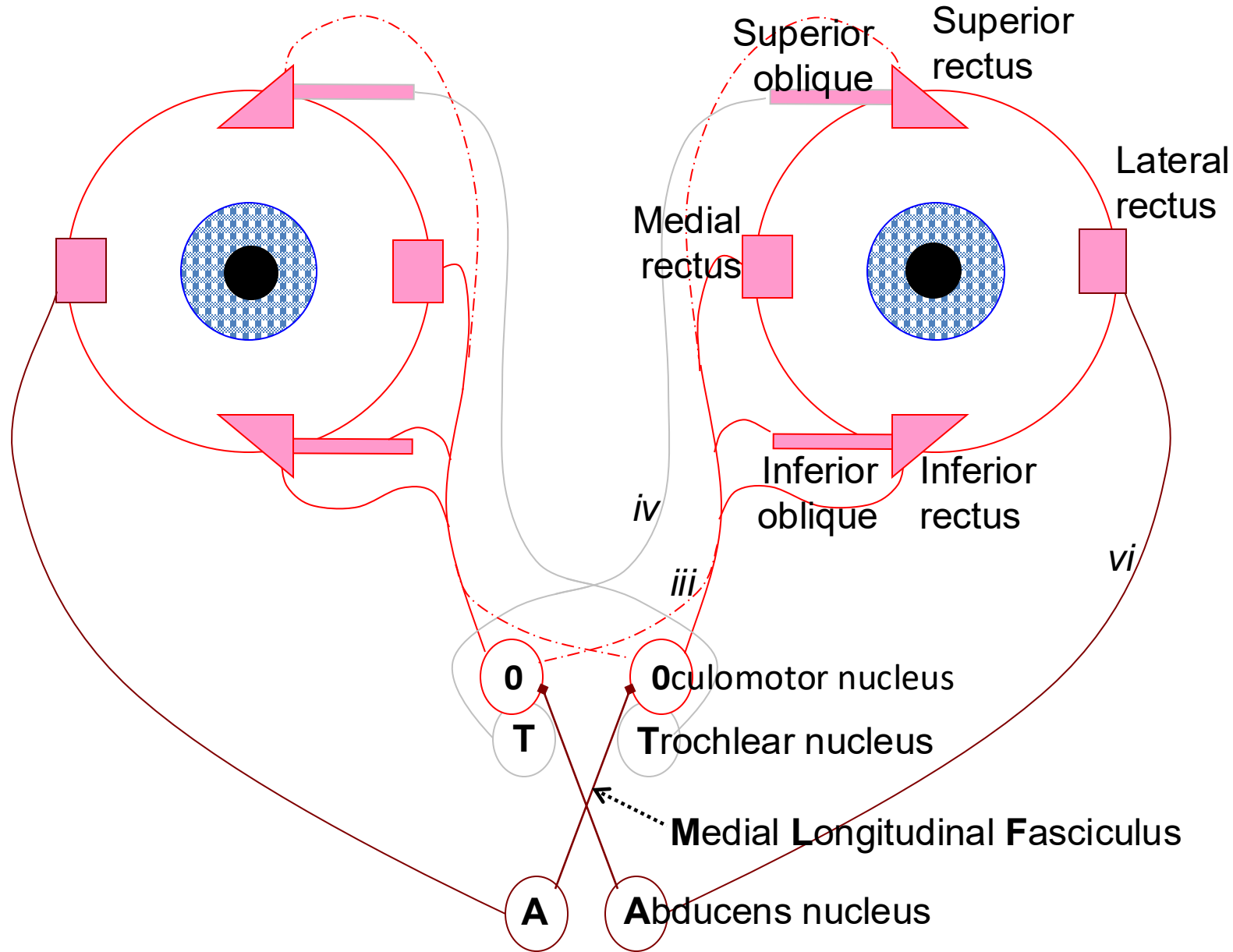
MLF

*Note: I said "medulla", but
should have said "midbrain".*

Abducens nucleus drives oculomotor nucleus via MLF

**NEW YORK INSTITUTE
OF TECHNOLOGY**

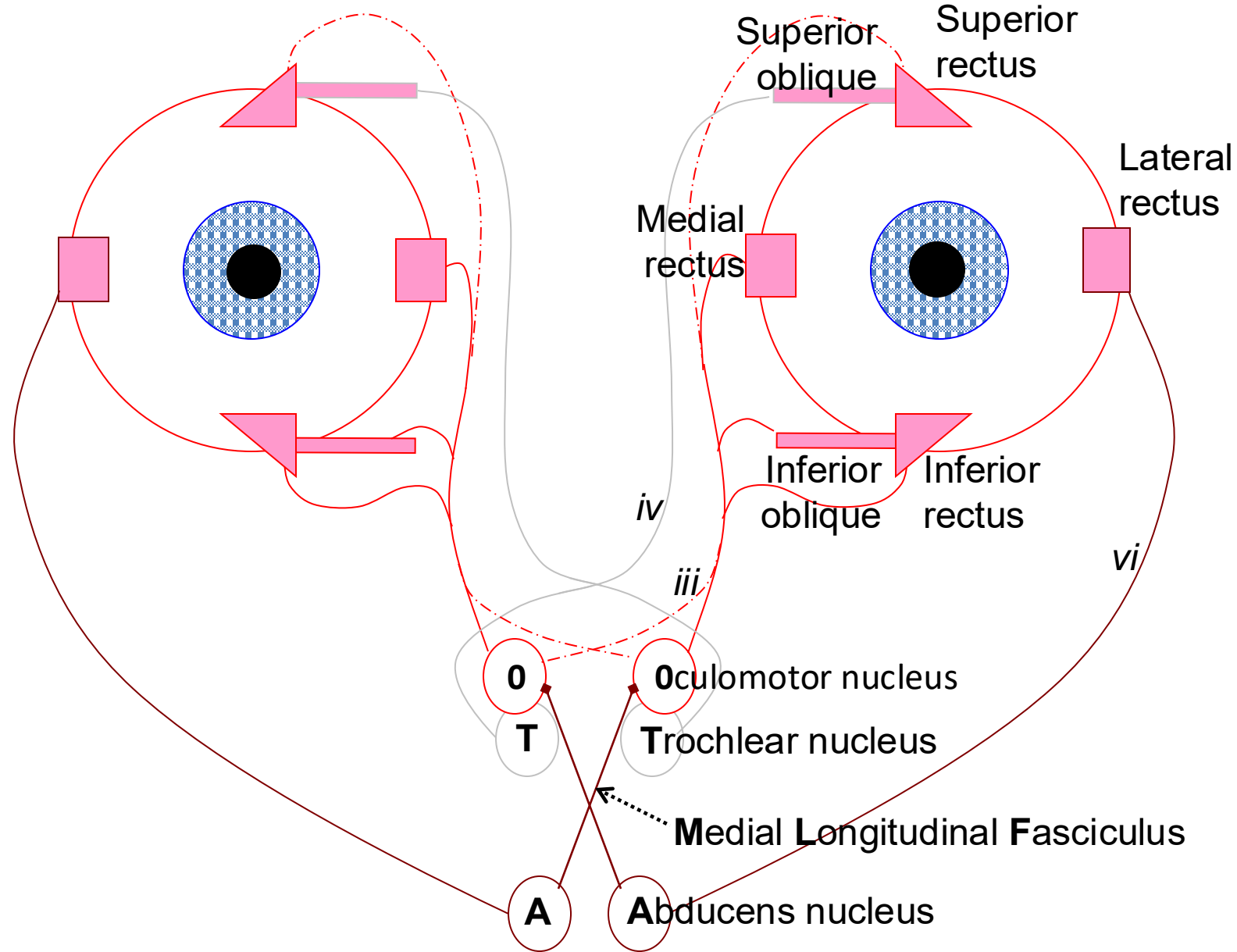
College of Osteopathic
Medicine



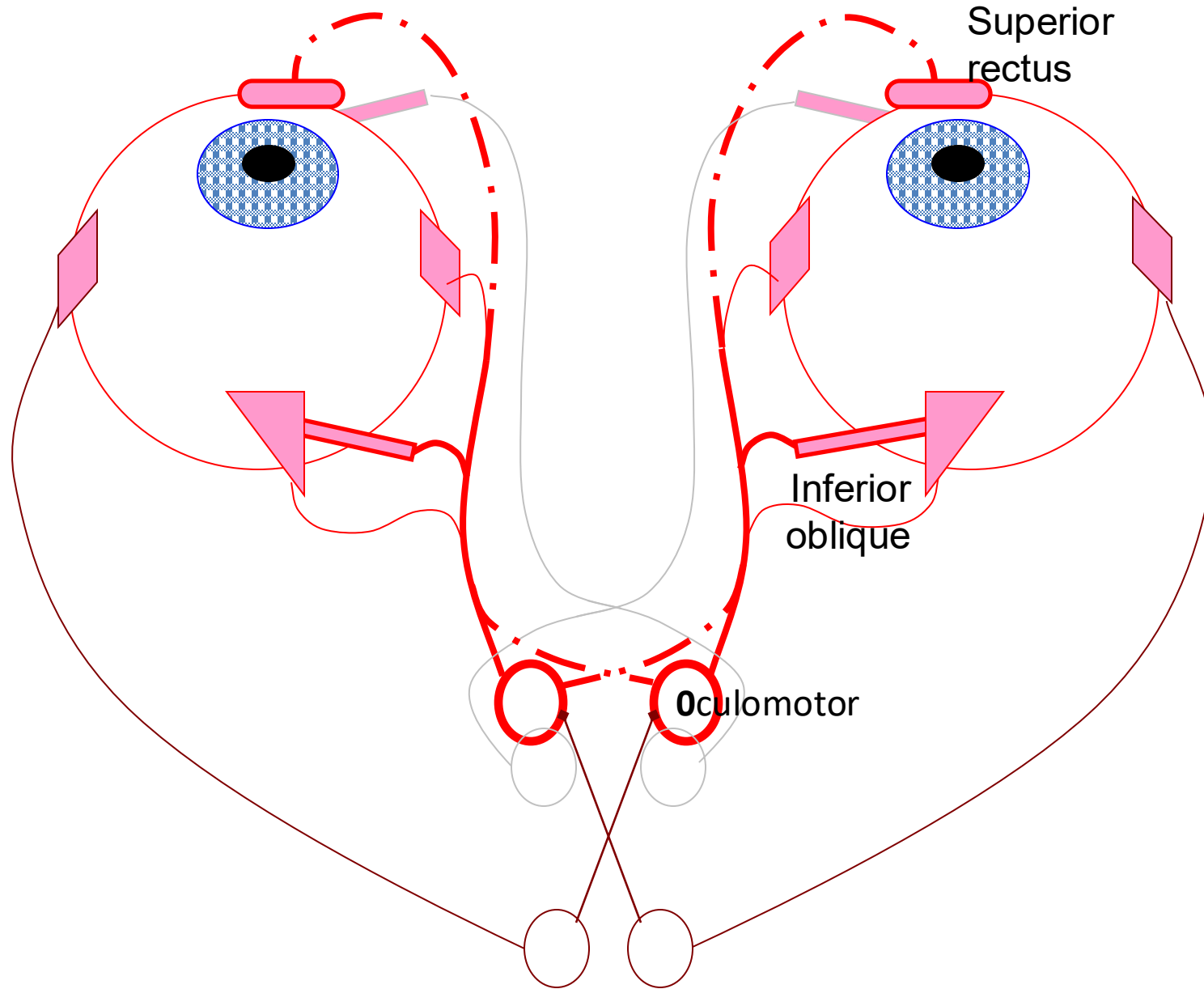
CN III, IV, VI control eye movements

**NEW YORK INSTITUTE
OF TECHNOLOGY**

College of Osteopathic
Medicine



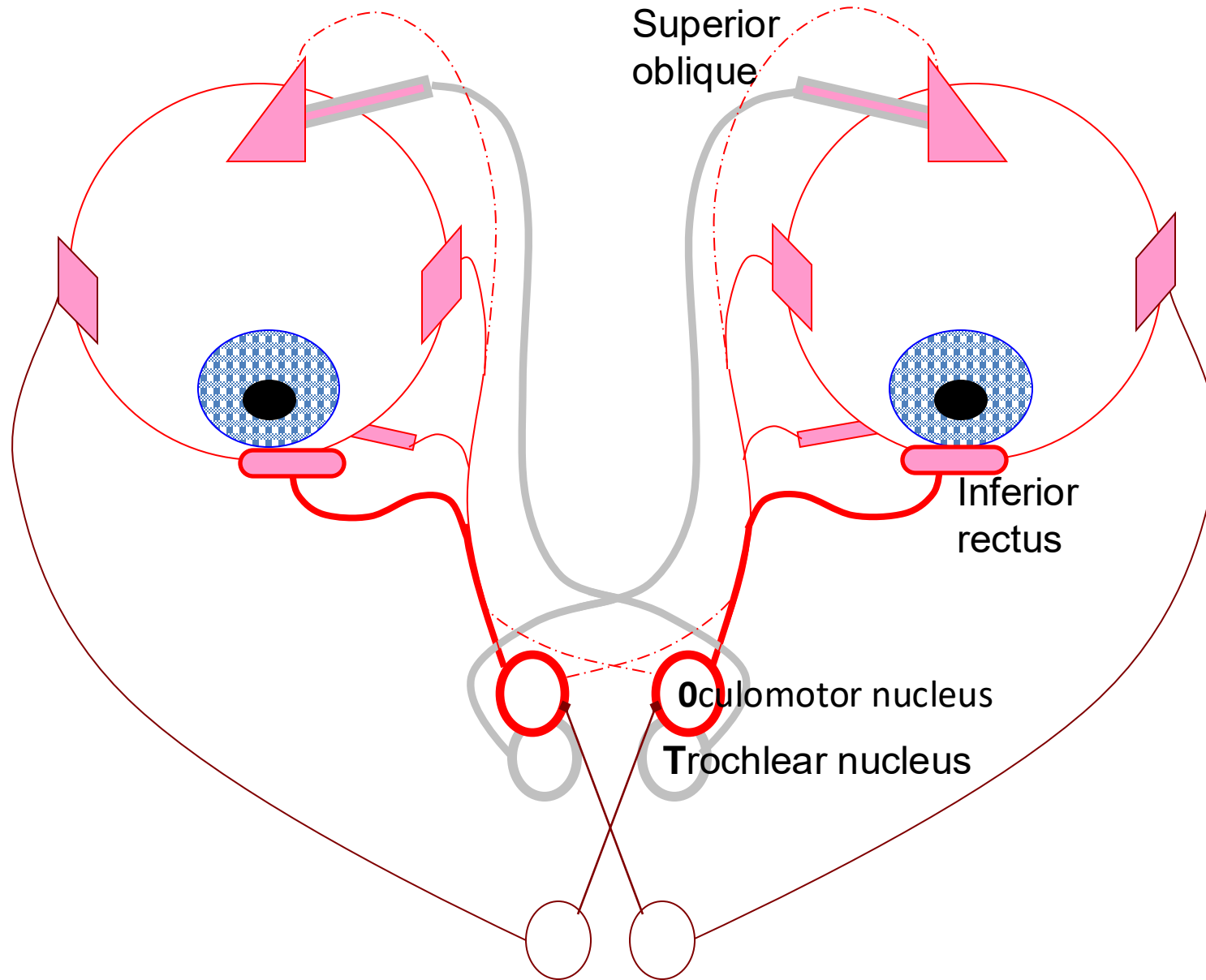
Elevation by CN III



**NEW YORK INSTITUTE
OF TECHNOLOGY**

College of Osteopathic
Medicine

Depression by CN III & IV



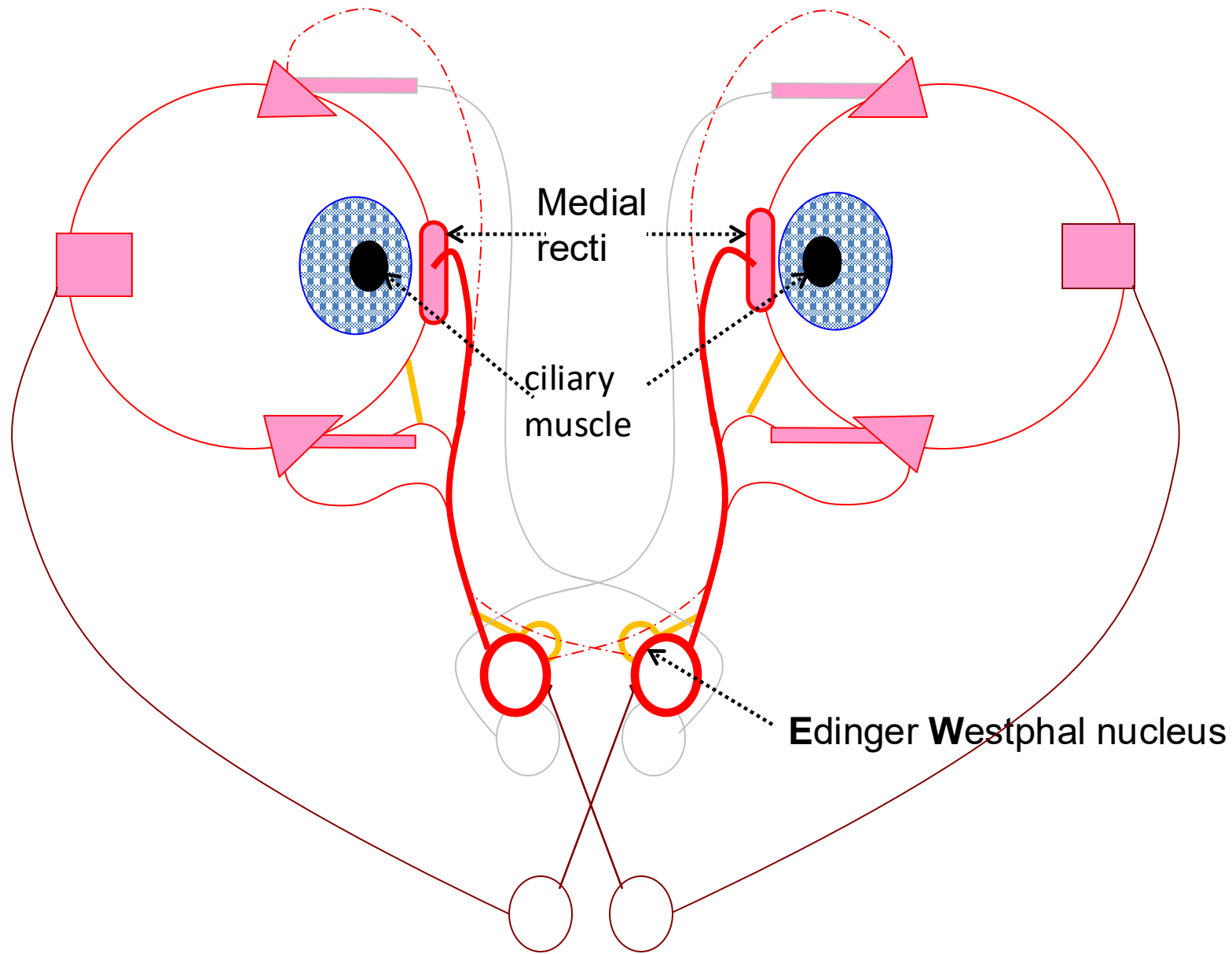
**NEW YORK INSTITUTE
OF TECHNOLOGY**

College of Osteopathic
Medicine

Vergence and accommodation by CN III

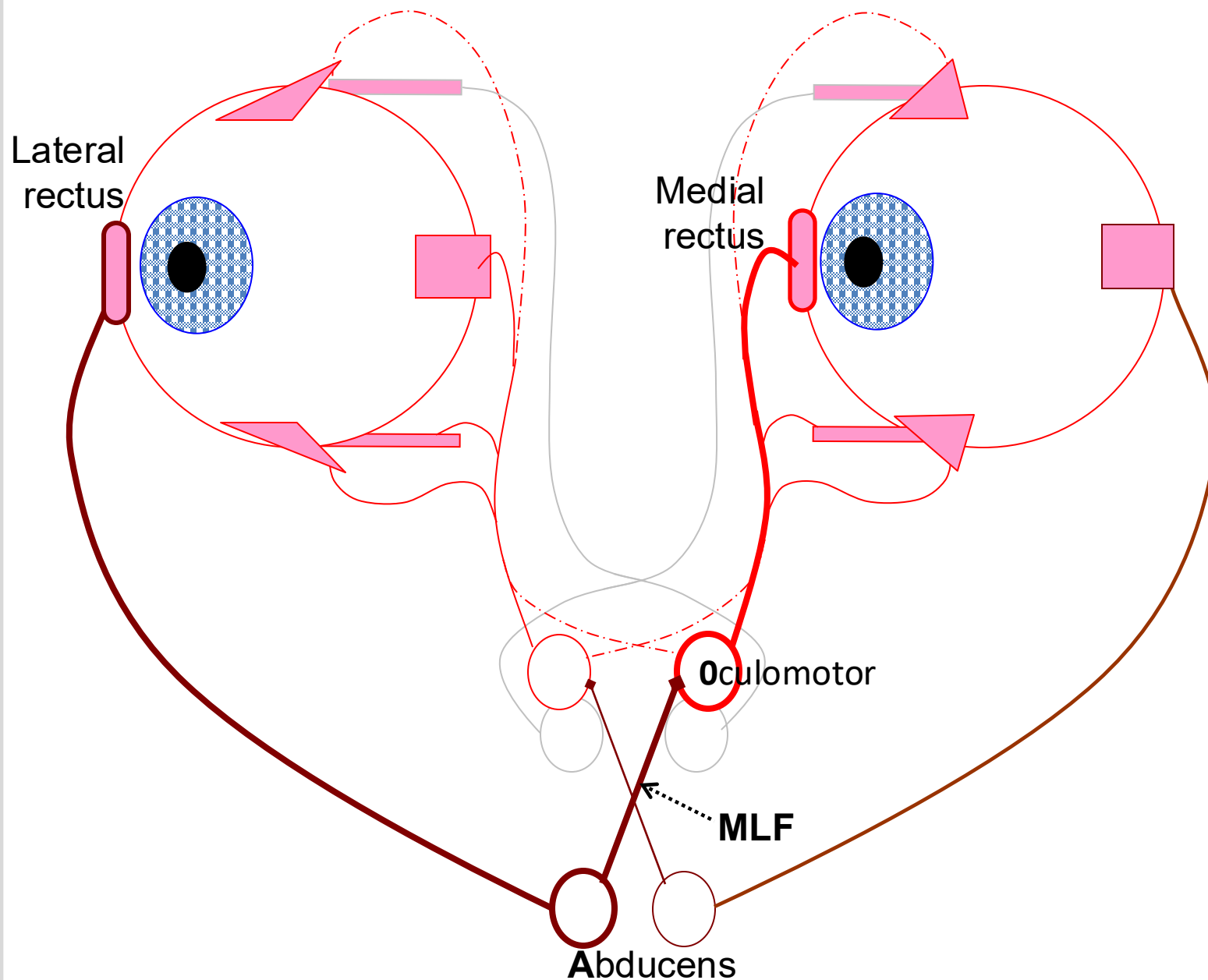
**NEW YORK INSTITUTE
OF TECHNOLOGY**

College of Osteopathic
Medicine



*Note: I said "vis-a-vis", but
should have said "via".*

Conjugate lateral gaze controlled by abducens & oculomotor nuclei



**NEW YORK INSTITUTE
OF TECHNOLOGY**

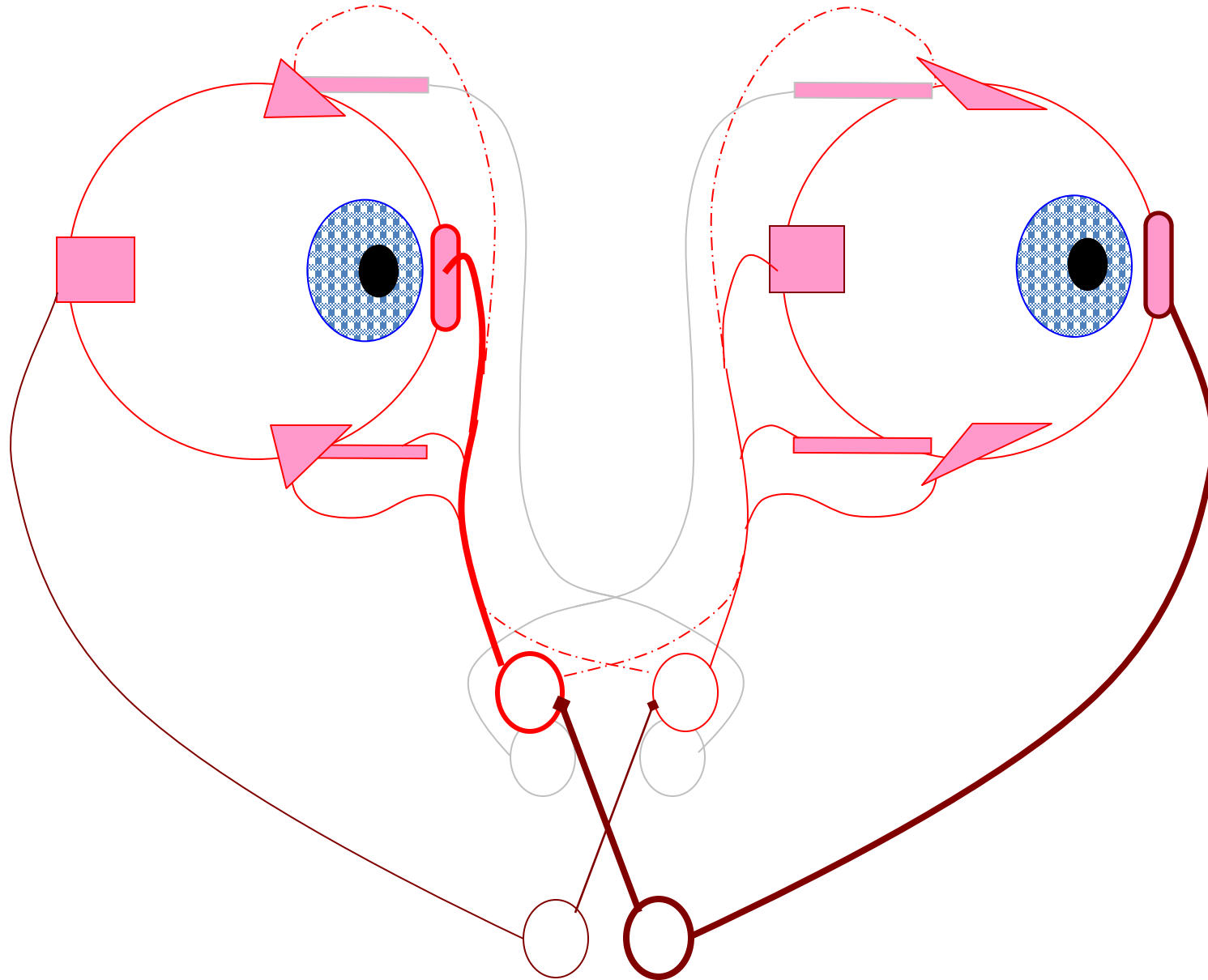
College of Osteopathic
Medicine

*Note: I said "vis-a-vis", but
should have said "via".*

Conjugate lateral gaze controlled by abducens & oculomotor nuclei

**NEW YORK INSTITUTE
OF TECHNOLOGY**

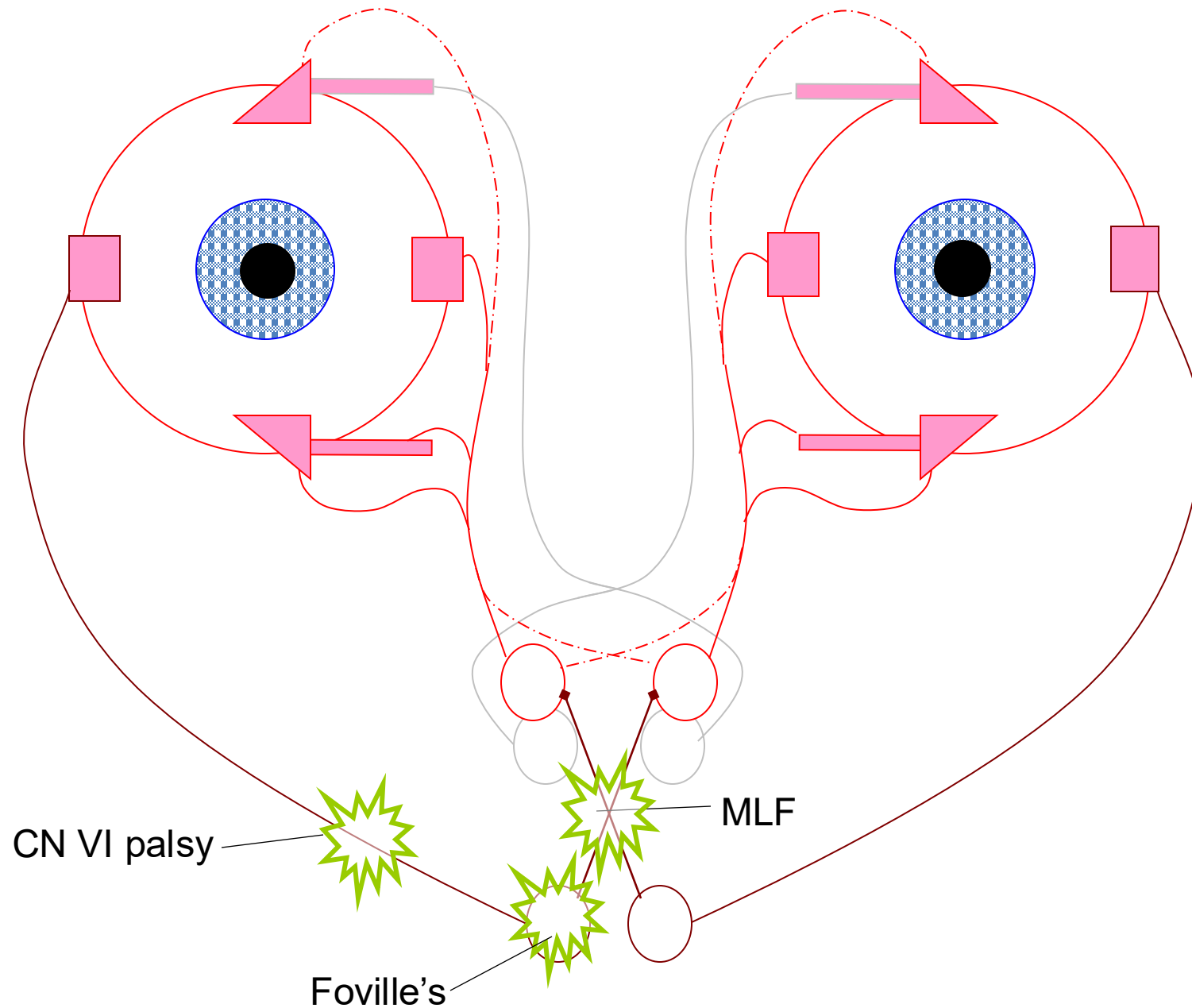
College of Osteopathic
Medicine



Lateral gaze dysfunction: CN VI, Foville's, & MLF syndromes

**NEW YORK INSTITUTE
OF TECHNOLOGY**

College of Osteopathic
Medicine

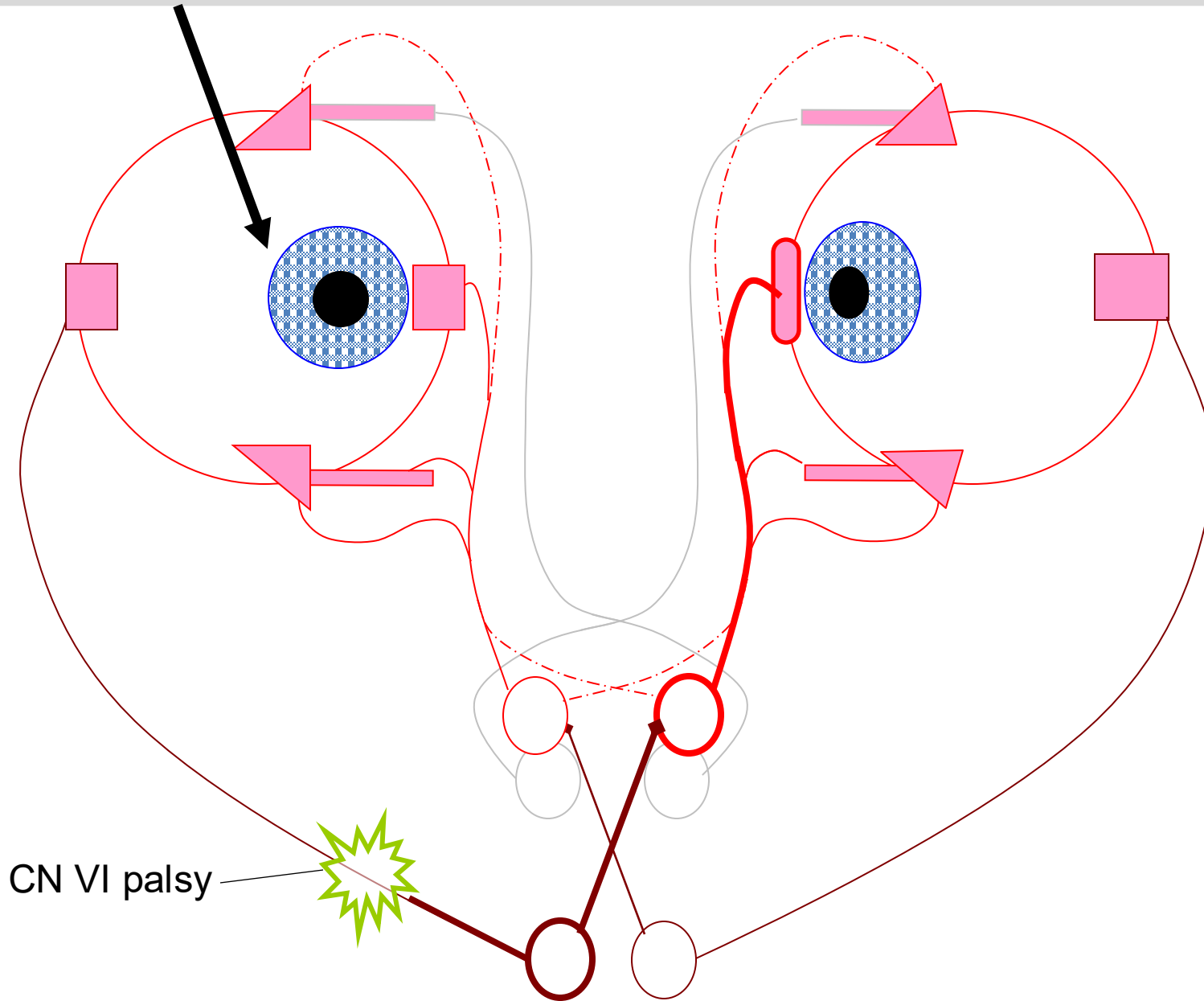


Whether gazing ahead or to either side

NEW YORK INSTITUTE OF TECHNOLOGY

College of Osteopathic
Medicine

affects only CN VI, but NOT
the MLF,
so contralateral adduction is
not affected

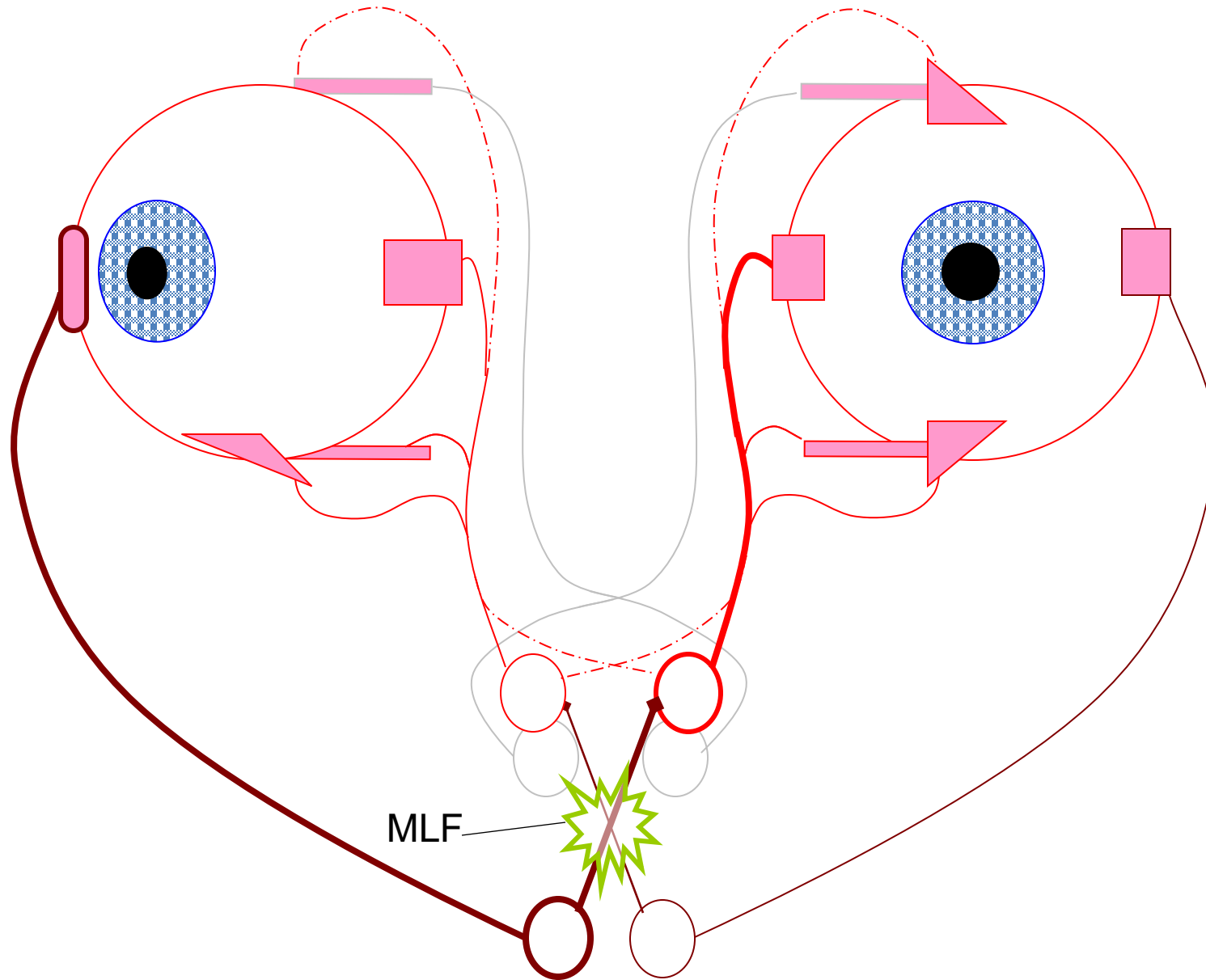


NEW YORK INSTITUTE OF TECHNOLOGY

College of Osteopathic
Medicine

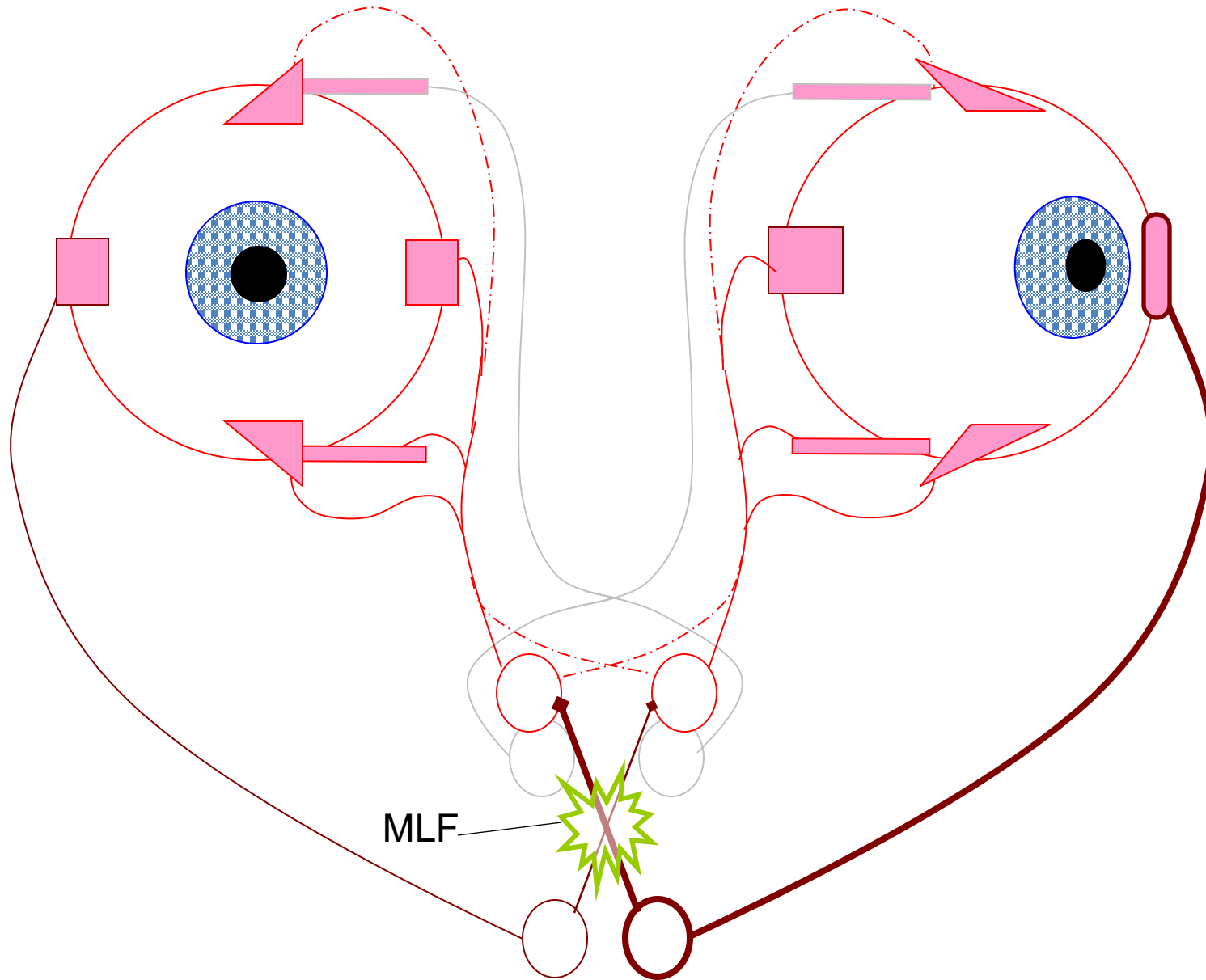
**MLF syndrome aka
internuclear
ophthalmoplegia:** affects
medial movement in the
context of conjugate
horizontal gaze

Remember, MLF is
bilateral and decussations
occur immediately before
entering. This example
depicts a lesion to both
columns.



NEW YORK INSTITUTE OF TECHNOLOGY

College of Osteopathic
Medicine

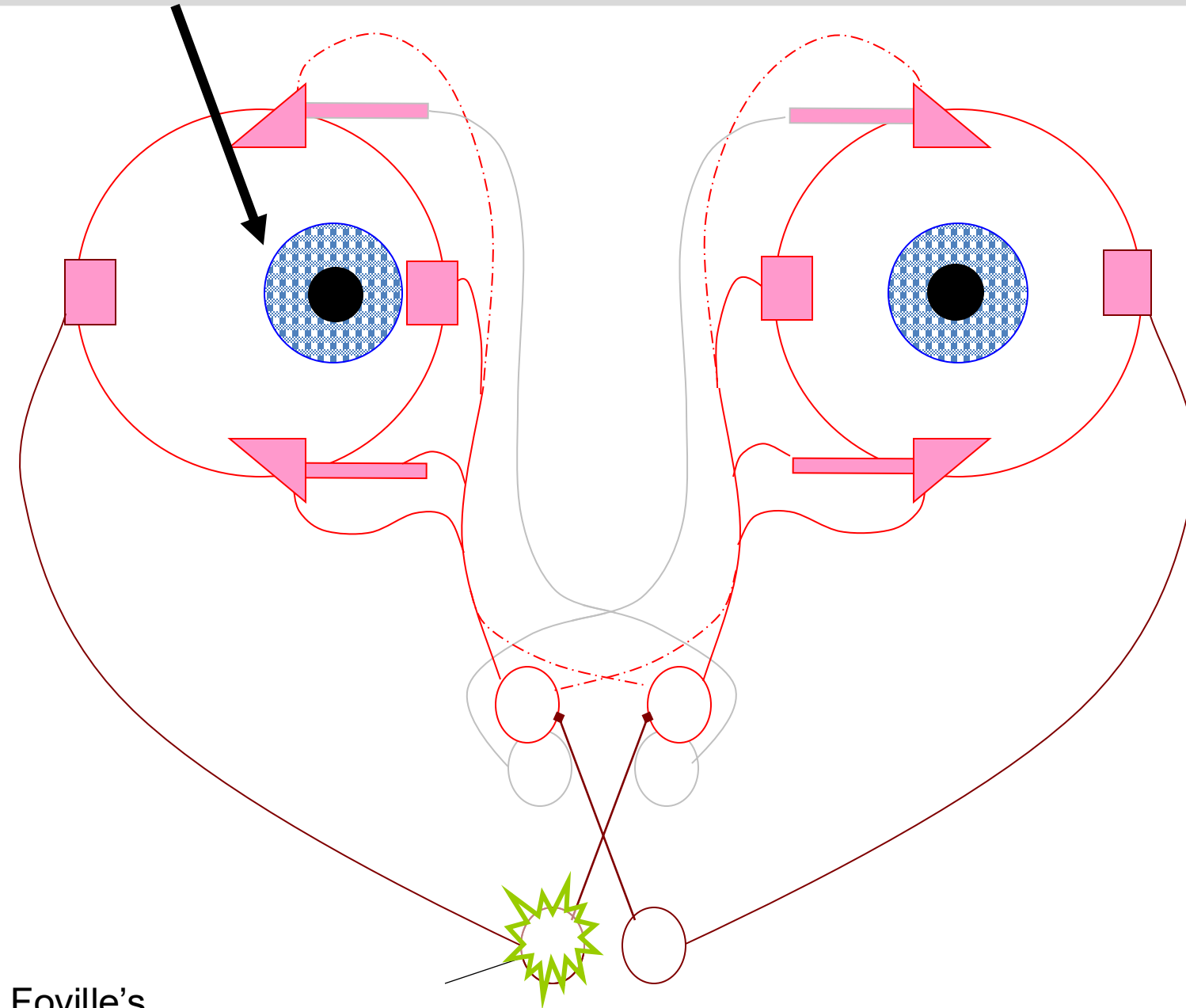


Whether gazing ahead or to either side

NEW YORK INSTITUTE OF TECHNOLOGY

College of Osteopathic
Medicine

Foville's syndrome knocks
out abducens nucleus,
impairing lateral eye
movement **toward** side of
lesion in BOTH eyes



Foville's
impaired conjugate gaze (and contralateral hemiparesis)

Summary

- Smell is conveyed by the olfactory tract.
- Vision is conveyed by optic tract, projects thalamus, superior colliculus, and pretectal area.
- Pretectal area bilaterally innervates the Edinger-Westphal nuclei and enables “consensual” light reflex.
- The mechanical action of the different eye muscles requires coordination among the vertical recti and opposing obliques.
- The oculomotor nucleus controls most eye muscles (4 of 6): the contralateral superior rectus, and the ipsilateral medial rectus, inferior rectus, and inferior oblique.
- The superior branch of the oculomotor nerve is from contralateral nucleus for the superior rectus and eyelid. Its inferior branch is from the ipsilateral nucleus for the other eye muscles, pupil and lens.
- The abducens nucleus controls the ipsilateral lateral rectus and contralateral oculomotor nucleus.
- Conjugate lateral gaze requires the medial longitudinal fasciculus.
- Trochlear nucleus controls the contralateral superior oblique.

Feedback is appreciated!

<https://comresearchdata.nyit.edu/redcap/surveys/?s=HRCY448FWYXREL4R>

